

Krüppel Regulates Cell Cycle Exit and Limits Adult Neurogenesis of Mushroom Body Neural Progenitors in *Drosophila*

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This study provides **important** insights into the regulation of neuroblast lifespan and proliferation in the *Drosophila* mushroom body, identifying Krüppel (Kr) as a key transcription factor promoting timely termination of these neuroblasts by repressing Imp expression, and proposes an antagonistic role of Krüppel homolog 1 (Kr-h1), whose overexpression leads to prolonged mushroom body neuroblast proliferation and tumor-like expansion. The findings are impactful for researchers interested in temporal patterning and neural development, and the methods and data analysis are **solid**, however, the precise regulatory interactions between Kr and Kr-h1 and their modes of action remain **incompletely** tested. Further experiments would be required to fully elucidate the mechanistic interplay between the factors involved.

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Abstract

In many organisms, including *Drosophila* and humans, neural progenitors exit the cell cycle and are eliminated by the end of development, which restricts adult neurogenesis to specific brain regions such as the mammalian hippocampus. Here, we show that the conserved transcription factor Krüppel (Kr) regulates the proliferation and neurogenic capacity of mushroom body neuroblasts (MBNBs), which generate the learning and memory center in the *Drosophila* brain, functionally analogous to the hippocampus. Neuroblast-specific *Kr* RNAi and the *Irregular facet* (*Kr^{If-1}*) mutation extends MBNB lifespan, enabling continued neurogenesis in the adult brain. Kr is expressed at low levels in postembryonic MBNBs, and its pupal stage-specific depletion is sufficient to induce MBNB retention, distinguishing this role from its established function in embryonic neurogenesis. Persisting MBNBs maintain expression of the RNA-binding protein IGF-2-binding protein (Imp), which promotes MBNB proliferation and early neuronal fate. Co-depletion of Imp abolishes extended neurogenesis induced by Kr depletion. Additionally, Krüppel homolog 1 (Kr-h1), another Kr family protein and a key regulator of hormone-mediated transcription, antagonises Kr's function: its

knockdown suppresses the *Kr* depletion phenotype while its overexpression drives tumour-like neuroblast overgrowths. These findings define a lineage-specific regulatory axis governing adult neurogenesis in *Drosophila*, with potential parallels in other organisms.

Introduction

The adult brain is predominantly composed of postmitotic neurons, which lack the capacity for proliferation. This limitation makes brain tissue particularly vulnerable to damage or degeneration, as lost neurons are largely irreplaceable. Consequently, neuronal loss significantly contributes to the decline of brain function, underlying various chronic pathologies and degenerative diseases associated with aging, such as Alzheimer's and Parkinson's diseases (1,2). Although a small population of proliferative progenitor cells persists in the adult brain, these cells are sparse and primarily quiescent, significantly limiting the brain's overall regenerative capacity (1,3).

Neural progenitors are essential for brain development, generating the vast majority of the brain's cellular constituents. These cells typically undergo asymmetric cell divisions to self-renew and produce differentiated progeny that mature into neurons or glial cells, or remain as partially differentiated intermediate progenitors (4,5). However, the majority of neural progenitors are eliminated during the late stages of development, leaving behind only a small pool in specialised niches like the subgranular zone in the hippocampus. The remaining progenitors are usually slow-proliferating and dormant but capable of reactivation in response to physiological or pathological stimuli (1,3). While the elimination of neural progenitors is vital for preserving existing neural circuits, keeping a small subset of progenitors is necessary for repairing brain injury or refining and remodeling neural pathways that support key function during adulthood, such as learning and memory (3,6,7). The mechanisms enabling this region-specific control of neurogenic capacity, however, remain poorly understood.

Drosophila melanogaster serves as an exemplary model organism for studying the development and regulation of neural progenitors in the central nervous system. In *Drosophila*, neuroblasts (NBs) are responsible for generating virtually all cells in the fly brain. NBs predominantly undergo asymmetric cell division to generate a daughter NB and a ganglion mother cell, which further divides once to become neurons or glia (5,8). The neurogenic capacity of NBs, as well as their cell cycle control and survival, are tightly regulated in a lineage/region-specific manner during development (9,10). At the end of embryogenesis, while many NBs in the abdominal and thoracic segments undergo apoptosis, the majority in the central brain enter a phase of quiescence, reactivated only by larval hatching and subsequent food intake (8,11,12). An exception to this regulation are mushroom body neuroblasts (MBNBs), the neuroblast population responsible for forming the mushroom body, a critical structure in insect brains associated with learning and memory, akin to the function of the hippocampus in mammals (13,14). MBNBs continue dividing during the embryo-to-larva transition (9,15). Furthermore, unlike other NBs that cease dividing and undergo apoptosis or differentiation before adult eclosion, MBNBs persist significantly longer, maintaining their neurogenic functions into the late pupal stages (9,16,17). Notably, in some insects like the house cricket *Acheta domesticus* and the moth *Agrotis ipsilon*, MBNBs not only surpass their usual developmental limits but also maintain active neurogenesis throughout the adult phase (18,19).

Complex interactions between cell-intrinsic mechanisms and extracellular cues intricately control the neurogenic capacity of NBs. Temporal transcription factors (tTFs) expressed in neuroblasts (NBs) in a sequential cascade during development confer temporal identity essential for generating the diversity of neural lineages in the central nervous system, exemplified by the well-characterised sequence of Hunchback (Hb), Krüppel (Kr), Pdm (Nubbin/Pou domain), Castor (Cas), and Grainyhead (Grh) in embryonic thoracic and many Type I NB lineages (20–22). A similar

progression of a series of transcription factors is also observed in mammalian brain development (23 [↗](#), 24 [↗](#)). In addition to providing temporal identity, tTFs also play a pivotal role in regulating NB proliferation and cell cycle progression during development. Early tTFs, such as Hb and Kr, promote NB proliferation, while later factors, including Cas and Svp, contribute to limiting NB divisions and eventually triggering cell cycle exit (11 [↗](#), 25 [↗](#)). The transition from early to late tTFs is crucial for determining the length of the neurogenic period and the timing of NB termination (25 [↗](#), 26 [↗](#)). In particular, Cas and Svp extend their influence into postembryonic stages, where they regulate neurogenesis termination by modulating the expression of key cell cycle regulators. Experimental manipulation of these factors, such as Cas overexpression or svp mutations, has been shown to prolong NB proliferation, allowing them to persist and continue generating neural progeny into adulthood (11 [↗](#), 27 [↗](#), 28 [↗](#)). Furthermore, RNA-binding proteins, Imp (IGF-II mRNA Binding Protein) and Syp (Syncrin), regulate the timing of NB cell cycle exit and differentiation during postembryonic development. These proteins are expressed in opposing gradients, where Imp promotes NB proliferation and early neural fate and Syp facilitates their transition to late fate and neurogenesis termination (29 [↗](#), 30 [↗](#)). While the Imp-Syp gradient is a common regulatory mechanism in many NBs, the regulation of this gradient varies across NB lineages. In Type I NBs, the postembryonic tTF Svp promotes the Imp-to-Syp transition through the activation of ecdysone signaling via induction of the Ecdysone Receptor (EcR) (27 [↗](#), 28 [↗](#)). In contrast, in MBNBs, where the transition is delayed, extracellular signaling through the activin receptor Baboon, activated by Myoglianin from surrounding glial cells, promotes this delayed transition (31 [↗](#)). Additionally, Ecdysone-induced protein 93F (E93), a transcription factor that mediates various actions of ecdysone signaling during metamorphosis, has been shown to specifically promote the termination of MBNBs by enhancing autophagy at late pupal stages (32 [↗](#)). However, the full range of regulatory mechanisms specific to MBNBs remains unclear.

The Krüppel-like family of transcription factors (KLFs) are evolutionarily conserved across metazoans and play a pivotal role in development, physiology, and stem cell regulation (33 [↗](#), 34 [↗](#)). *Drosophila* Kr, a founding member of this family, was first identified as a gap gene, playing a crucial role in establishing the anterior-posterior axis during embryonic development (35 [↗](#)). While its critical role as a tTF in embryonic neurogenesis has been recognised (25 [↗](#)), the involvement of Kr in postembryonic neurogenesis and beyond has not been unexplored. Another KLF member, Krüppel homologue 1 (Kr-h1), is recognised as a target gene of hormonal signalling (36 [↗](#)–38 [↗](#)). Kr-h1 has been implicated in neuron remodeling, particularly under the influence of ecdysone during metamorphosis, suggesting its involvement in the dynamic regulation of neurogenesis (39 [↗](#), 40 [↗](#)). However, the roles of Kr and Kr-h1 and their potential interactions in specific NB lineages remain uncharacterised.

In this study, we uncover a novel function of Kr in regulating the survival and proliferative capacity of MBNBs in *Drosophila* adult brains. We show that NB-specific depletion of Kr, as well as the classic *Irregular facet* (*Kr^{If-1}*) mutation, previously associated with morphological defects in adult eyes (41 [↗](#), 42 [↗](#)), extends MBNB lifespan and maintains neurogenesis in the adult brain. *Kr^{If-1}* thus becomes the first spontaneous mutation identified that alters adult neurogenesis in metazoans. Additionally, we demonstrate that, distinct from its role as a tTF in embryonic NBs, Kr regulates the expression and function of the RNA-binding protein Imp to control the timely cell cycle exit of MBNBs during late pupal stages. Furthermore, we provide evidence for the potential antagonism between the two Kr family proteins, Kr and Kr-h1, in regulating MBNB cell cycle progression and neurogenic capacity. Our findings highlight the critical role of the conserved Kr family transcription factors in the longevity and neurogenic potential of specific neural progenitor pools, opening new avenues for understanding the genetic regulation of neurogenesis in adult brains.

Results

Postmitotic State in the *Drosophila* Adult Central Brain

Early studies concluded that no adult neurogenesis occurs in *Drosophila* adult brains (9,16,17). However, a subsequent study employing a highly sensitive lineage tracing technique demonstrated that slow neurogenesis continues beyond adulthood in optic lobes, which can be accelerated by acute injury (43). Additionally, quiescent progenitor cells have been identified in the central brain, capable of inducing neurogenesis or gliogenesis when stimulated by physical damage or excessive neuron death (44,45). However, under unperturbed conditions, no proliferating cells have been reported in the adult central brain.

To confirm this, we examined the presence of proliferating cells by immunostaining for the mitotic marker phosphorylated histone H3 Serine 10 (pH3) and by performing EdU incorporation assays to detect DNA synthesis. In control adult brains after five days of EdU feeding, we observed neither pH3-positive cells nor any EdU incorporation (Fig. 1A, Fig. 3A, B). We also examined cell cycle states in brain tissue using Fly-FUCCI, a fluorescent reporter system that visualises cell cycle phases via two differentially fluorescently labeled protein fragments that undergo cell cycle-dependent proteolysis: EGFP-E2F1 (amino acids 1-230) and mRFP-NLS-CycB (amino acid 1-266) (Fig. 1A) (46). Virtually all cells in the central brain showed only EGFP-E2F1 without mRFP-NLS-CycB signals, indicating that they were in G1 or G0 phase (Fig. 1A). Occasionally, one or two cells near the antennal lobes displayed both EGFP and mRFP signals, indicative of G2 phase (Fig. 1A). These cells may correspond to G2 cells previously reported in the adult brain during the so-called “critical period,” which spans five days from eclosion (7,47).

To gain insight into the molecular mechanisms maintaining the postmitotic state of the adult brain, we analysed the expression patterns of cell cycle regulator (CCR) genes using recent single-cell RNA sequencing (scRNAseq) data from *Drosophila* adult and larval brains (48,49). The sequencing data for adult and larval brains were obtained, and cells were organised into 17, 84, or 29 distinct clusters, respectively, based on their gene expression profiles (Fig. 1B, C, S1). We then analysed the transcription levels of 112 selected CCRs in each cluster (Fig. 1B, C, S1).

Most positive cell cycle regulators (CCRs), which promote cell cycle transitions, including *Cdk1/2/4/7*, their partner cyclins, mitotic kinases Polo and Aurora A/B, and components of the replication machinery, were found to be minimally expressed across all neuronal and glial cell clusters in both adult and larval brains, while being highly expressed in NB clusters in larval brains (Fig. 1B, C, S1). However, an exception was observed for E2F1-Dp, a transcriptional activator complex driving the G1/S transition, as well as Myc and Max, a heterodimeric transcription factor that promotes both cell proliferation and growth through transcriptional regulation. These two complexes exhibited relatively high expression levels in various neuronal and glial clusters (Fig. 1B, C, S1). In mammals, E2F1-Dp has been implicated in regulating neuronal migration and apoptosis (50–52), while Myc has been associated with neuronal differentiation and activity (53,54). These findings suggest that E2F1-Dp and Myc-Max may serve neuron-specific functions beyond cell cycle regulation in *Drosophila*.

In contrast, negative CCRs, which function to delay or block cell cycle progression, often by inhibiting positive CCRs, remained relatively high across both NB and neuronal clusters (1B, C, S1). These include components of the anaphase promoting complex/cyclosome (APC/C), such as *fizzy-related* (*fzr*, the *Drosophila* FZR1 orthologue), *vihar* (*vih*, UBE2C), CG15237 (APC15), and CG8188 (Ube2S), as well as the DNA licensing inhibitor *geminin* and the checkpoint kinase *grape* (*grp*, the CHK1 orthologue),

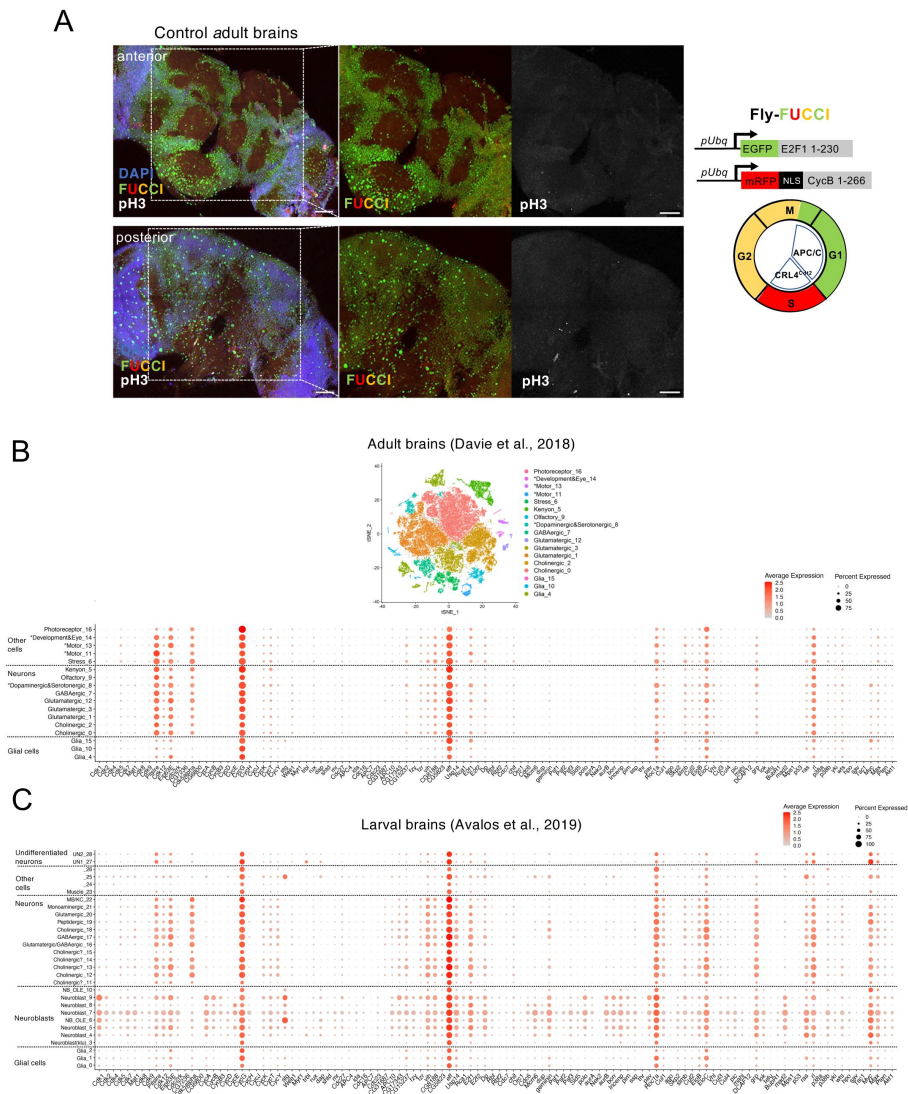


Figure 1

The *Drosophila* adult central brains are post-mitotic

(A) (Left panels) Representative confocal images of control (w^{1118}) adult central brains ubiquitously expressing the Fly-FUCCI cell cycle reporters, which label cells in distinct phases of the cell cycle. GFP-tagged E2F1 amino acid 1-230 fragment (EGFP-E2F1, green) marks cells in G1/G0 phase, while RFP-tagged Cyclin B amino acid 1-266 fragment (mRFP-NLS-CycB, red) marks cells in S/G2/M phases. Brains were counterstained with DAPI (blue) for DNA and phosphorylated histone H3 (pH3, grey) to label mitotic cells. Upper panels show anterior views, and lower panels show posterior views. In the adult central brain, nearly all cells are GFP-positive and lack RFP signals, confirming that they reside in the G1/G0 phase. No pH3-positive mitotic cells were detected, indicating that neurogenesis does not occur under normal conditions. A small number of cells near the antennal lobes exhibit weak mRFP-NLS-CycB signals, suggesting rare G2-phase cells. Scale bars: 50 μ m. (Right) Schematic diagram of the Fly-FUCCI system, illustrating its ability to label cell cycle states based on the differential degradation of GFP-E2F1 and RFP-CycB. **(B)** (Top) tSNE plot depicting the clustering of scRNAseq data from adult *Drosophila* brains (Davie et al., 2018). Cells are classified into 17 distinct clusters, with different colours representing different neuronal, glial, and neuroblast populations. (Bottom) Dot plot showing the expression of 112 selected CCR genes across the identified clusters. The colour intensity represents the average expression level, with darker red indicating higher expression, and dot size indicating the proportion of cells within each cluster expressing the respective gene. **(C)** Dot plot showing the expression patterns of CCR genes across 29 cell clusters identified from the single-cell RNA sequencing dataset of *Drosophila* larval brains (Avalos et al., 2019). Colour intensity and dot size are as described in (B). These analyses reveal that most positive CCRs are highly expressed in neuroblast clusters in larval brains but are minimally expressed in adult neuronal and glial clusters, supporting the postmitotic state of the adult brain.

Besides these canonical CCRs, non-canonical CDKs that function in transcriptional regulation in addition to cell cycle control (55 [↗](#)), such as Eip63E/CDK10, Pitslre/Cdk11, and Cdk12, along with their partners CycG, CycK, CycT, and CycY, were relatively highly expressed across both neural and NB clusters (Fig. 1B [↗](#), C, S1 [↗](#)). Similarly, several CCRs involved in ubiquitin-dependent protein degradation, such as an E2 ubiquitin conjugating enzyme *effete* (*eff*, the *Drosophila* Ube2d2 orthologue), as well as components of Cullin-RING-type E3 ligase complexes, APC/C, SCF, and CRL2, and the deubiquitinase Usp14, exhibited modest to high expression across neural and NB clusters (Fig. 1B [↗](#), C, S1 [↗](#)), underscoring the importance of the ubiquitin-proteasome pathway in neuronal regulation (56 [↗](#)). A particularly notable CCR was the Cdk5 partner Cdk5a (mammalian p35 orthologue), which exhibited significantly higher expression in neural clusters than in NB clusters, supporting its established role in postmitotic neurons (57 [↗](#)) (Fig. 1B, C [↗](#)). In summary, our transcriptome analysis suggests that most CCRs primarily maintain the postmitotic state of the adult brain through their canonical functions, whereas a subset of CCRs may regulate specific neuronal and/or glial functions independently of cell cycle regulation.

The above results suggest that repression of positive CCRs, rather than upregulation of negative CCRs, is critical for maintaining the postmitotic state of adult brain tissue. We therefore directly tested whether forced expression of positive CCRs could drive postmitotic neurons into the cell cycle by overexpressing these CCRs in mature mushroom body neurons and dopaminergic neurons adult neurons using *mb247-Gal4* and *TH-Gal4*. Co-expression of E2F1-Dp and Cdk2-CycE, but not Cdk4-CycD or CycE and String/CDC25, was sufficient to induce mitotic entry in both neuron types when expression was induced in young adult flies (less than two days post-eclosion) (Fig. S2A [↗](#)). However, when E2F1-Dp and Cdk2-CycE were induced 10 days after eclosion, no mitotic neurons were observed ($n \geq 10$), suggesting that neurons become increasingly refractory to cell cycle entry with age. Notably, these mitotic neurons frequently exhibited Dcp-1 signals, indicative of apoptosis (Fig. S2B [↗](#)), suggesting that forced cell cycle entry may lead to DNA damage accumulation or mitotic defects in neurons. To assess the physiological impact of this aberrant mitotic entry, we conducted a survival analysis and found that adult flies with E2F1-Dp and Cdk2-CycE induction in dopaminergic neurons exhibited significantly shorter lifespans compared to controls (lacZ overexpression). The median lifespan of control flies was 22 days, whereas flies overexpressing E2F1-Dp and Cdk2-CycE had a significantly reduced median lifespan of 14 days (Fig. S2C [↗](#)). Together, these findings suggest that adult brain tissue actively represses positive CCRs to prevent unscheduled proliferation and maintain their postmitotic state.

The neuroblast-specific knockdown and mutation in the transcription factor *Krüppel* leads to the presence of mitotic cells in the adult central brain

Having confirmed the postmitotic state of the adult *Drosophila* central brain, our next objective was to identify the regulators that establish and maintain this condition, with a particular focus on NBs. Given the intricate regulatory network that governs NB activity, we hypothesised that specific transcription factors might be key players. To test this we performed a targeted genetic screen focused on conserved transcription factors known for their roles in neurogenesis control and stem cell regulation.

Through this screen, *Krüppel* (Kr) emerged as a strong candidate for regulating NB cell cycle exit and/or quiescence. Strikingly, when we depleted Kr specifically in NBs using the *insc-Gal4* driver to induce RNAi-mediated knockdown, we observed the presence of proliferative clones, identified by EdU incorporation, in the dorsoposterior region of the central brains in adult flies (Fig. 2A, B [↗](#)). Two independent *Kr* RNAi constructs (v104150 and v40871 from the Vienna *Drosophila* Resource Center, hereinafter referred to as *KrIR#1* and *KrIR#2*, respectively) exhibited a comparable phenotype, confirming that this effect was specific to Kr depletion rather than an off-target effect (Fig. 2A, B [↗](#)).

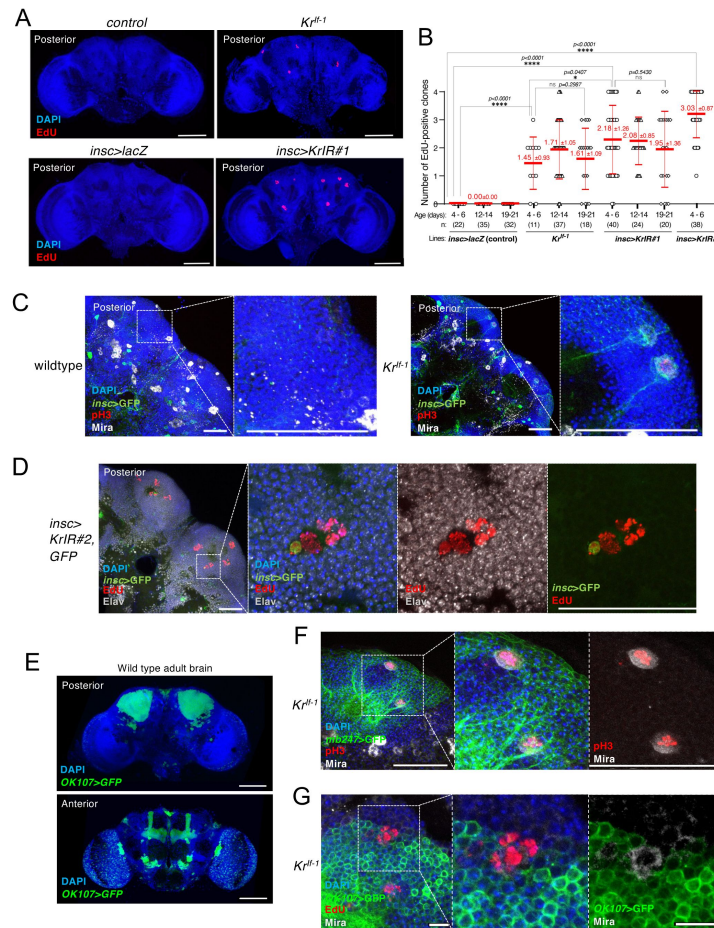


Figure 2.

Neuroblast-specific Kr depletion and *Kr^{If-1}* mutation causes mushroom body neuroblast retention and prolonged neurogenesis in adult brains.

(A) Representative images of EdU-labeled proliferative clones in adult central brains. Adult flies of control *Kr* wild type (*Kr⁺/Kr⁺*, upper left panel), heterozygous *Kr^{If-1}* mutant (*Kr^{If-1}/Kr⁺*, upper right panel), UAS control *insc>lacZ* (lower left panel), and *KrRNAi insc>KrIR#1* (lower right panel) were subjected to EdU incorporation assays and stained with DAPI (blue) and EdU (red). In both control brains, no EdU-positive clones were detected. In contrast, *insc>KrIR#1* and *Kr^{If-1}* mutant brains exhibited multiple EdU-positive clones per hemisphere, predominantly in the dorsoposterior region. Scale bars: 100 μ m. **(B)** Quantification of EdU-positive clones in adult brains aged 4–6 days, 12–14 days, and 19–21 days post-eclosion. Scatter dot plots represent the number of EdU-positive clones per hemisphere. The means and standard deviations are depicted by thick and thin red bars, respectively, with actual values annotated above. Statistical significance was determined using the Mann-Whitney U test (*: $p \leq 0.05$, **: $p \leq 0.01$, ***: $p \leq 0.001$, ****: $p < 0.0001$, ns = non-significant). 'n' represents the number of brain hemispheres analysed per condition. **(C)** *Kr* wild type (left) and *Kr^{If-1}* mutant (right) adult brains expressing *mCD8::GFP* (green) under the NB-specific *insc*-Gal4 driver were stained with DAPI (blue), pH3 (red), and Mira (gray). Mitotic and interphase NBs, identified by co-expression of Mira and GFP, were detected in the dorsoposterior regions of *Kr^{If-1}* mutant brains but not in controls. Scale bars: 50 μ m. **(D)** *insc>KrIR#2* adult brains co-expressing *mCD8::GFP* were stained with DAPI (blue), EdU (red), and the pan-neuronal marker Elav (gray). EdU incorporation was observed in Elav-positive neurons, indicating continued neurogenesis in adult brains upon Kr depletion. Scale bars: 50 μ m. **(E)** Wild-type adult brain expressing *mCD8::GFP* (green) under the mushroom body (MB)-specific *OK107*-Gal4 driver to visualise the location and the structure of the MBs, stained with DAPI (blue). Posterior and anterior views are shown. The MB cell body region is located dorsoposteriorly, while MB lobes are visible in the anterior view. Scale bars: 100 μ m. **(F)** *Kr^{If-1}* mutant adult brains expressing *mCD8::GFP* driven by the MB-specific driver *mb247*-Gal4 (green), stained with DAPI (blue), pH3 (red), and Mira (grey). Mitotic cells in the MB cell body region exhibited cortical Mira localisation, characteristic of dividing NBs. Scale bars: 50 μ m. **(G)** Representative images of *Kr^{If-1}* mutant brains expressing *mCD8::GFP* under *OK107*-Gal4 after EdU labelling. EdU incorporation was observed in MBNBs, weakly marked by *OK107>GFP* expression and Mira staining. Scale bars: 10 μ m.

To further validate the link between *Kr* and the adult neurogenic phenotype, we examined various *Kr* mutant alleles. Notably, the *Irregular facet* (*Kr^{If-1}*) mutation, a dominant mutation in the *Kr* locus known to cause *Kr* misexpression in various tissues, including the eye imaginal discs (41 [42](#)), also exhibited EdU incorporation in multiple clones in adult brains, although at a somewhat lower frequency than *Kr* RNAi (Fig. 2B, C). These findings suggest that both loss of *Kr* function (via RNAi) and its misregulation (via *Kr^{If-1}* mutation) lead to the persistence of proliferative cells in the adult central brain.

The mitotic clones contain *bona fide* neuroblasts undergoing asymmetric division

To determine the identity of these proliferative clones, we analysed the expression of NB markers. As expected, the dividing cells expressed mCD8::GFP (under *insc*-Gal4 control) and the NB marker Miranda (Mira), confirming their NB identity (Fig. 2C, D). Each clone typically contained only one to two pH3-positive cells, consistent with the presence of actively cycling NBs within these clusters (Fig. 2C). In addition to NBs, the EdU-positive clones also contained cells expressing the pan-neuronal marker Elav, indicating that these mitotic NBs generate neurons (Fig. 2D). Notably, these NBs exhibited a crescent-like asymmetric localisation of Mira at the cell cortex during mitosis, with smaller surrounding cells lacking NB markers (Fig. 2F). This asymmetric distribution suggests that these NBs undergo conventional asymmetric cell division, producing both a self-renewing NB and a differentiating daughter cell.

The retained neuroblasts are mushroom body neuroblasts that persist into adulthood

We further characterised these anomalous mitotic NBs in *Kr*-depleted or *Kr^{If-1}* mutant adult brains. While interphase NBs, marked by Mira and *insc*>mCD8::GFP but lacking pH3 signals, were observed in the dorsoposterior region of *insc*>*KrIR* and *Kr^{If-1}* mutant adult brains, neither interphase nor mitotic NBs were detected in the same region in control adult brains (Fig. 2C). This suggests that *Kr* depletion or misexpression leads to the retention or *de novo* formation of NBs, rather than the reactivation of quiescent NBs that may persist in wild-type brains.

The number and position of the persisting NBs within the adult brains were highly reminiscent of Mushroom Body Neuroblasts (MBNBs), which are normally eliminated during late pupal stages via apoptosis (9, 29, 32). In *insc*>*KrIR* or *Kr^{If-1}* mutant adult brains, we consistently observed one to four EdU-positive clones per hemisphere, located on the dorsoposterior surface, matching the typical positioning of MBNBs during development (9) (Fig. 2A-F). To further verify their identity, we used *mb247*-Gal4, which labels MB neurons and found that the mitotic NBs were positioned within the MB cell body region (Fig. 2E, F). We then employed *OK107*-Gal4, which can label both MB neurons and MBNBs, and found that a subset of EdU-positive clones co-expressed *OK107*>GFP and Mira, confirming their MBNB identity (Fig. 2G).

Intriguingly, in both *insc*>*KrIR* and *Kr^{If-1}* mutant adult brains, we detected EdU-incorporated clones beyond 21 days after adult eclosion, far exceeding the previously described critical period of early adulthood during which NBs have been reported to retain transient mitotic activity (Technau, 2007; Li and Hidalgo, 2020) (Fig. 2B). This suggests that the observed phenotype is not merely a delay in MBNB elimination but rather represents a fundamental shift in MBNB fate, allowing their continued proliferation well into adulthood.

Thus, *Kr* depletion or its misexpression leads to the persistence of actively dividing MBNBs, which continue to generate neurons that integrate into the adult mushroom bodies. These findings indicate that *Kr* is essential for the timely elimination of MBNBs during pupal development, and its perturbation overrides this process, extending their neurogenic potential into adulthood.

Kr function in pupal MBNBs to control its cell cycle exit and elimination

Given these observations, we next sought to determine how Kr regulates MBNB elimination and whether it functions at a specific developmental stage and in a particular neuroblast lineage. Although Kr was initially characteriaed as a gap gene (35 [↗](#)), it has since been implicated in various cellular processes across different developmental stages, including neurogenesis. In Type I NBs, particularly those in the embryonic thorax, Kr acts as one of tTFs that regulate neural progeny identity and cell cycle dynamics (25 [↗](#)). Manipulations of tTF expression have been shown to alter the proliferation and survival of NBs, in some cases extending their lifespan (11 [↗](#), 21 [↗](#)). While genetic manipulation of Kr alone has not been reported to induce NB proliferation in adulthood, previous studies have shown that overexpression of Kr in combination with three other transcription factors can promote prolonged neurogenesis beyond developmental stages (26 [↗](#)). However, whether Kr acts as a tTF in MBNBs remains unclear. Unlike in other regions of the developing *Drosophila* brain, rapidly progressing series of tTFs have not yet been described in MBNBs (Doe, 2017 [↗](#); Holguera and Desplan, 2018; Kohwi and Doe, 2013; Rossi et al., 2017). Given the unique features of MBNBs, including their distinct neurogenic pattern and neuron types, it is possible that Kr plays a different role in their regulation.

To determine whether MBNB retention in Kr-depleted brains results from a malfunction of Kr as a tTF, we first examined Kr expression across different developmental stages. To assess Kr expression in MBNBs, we used two previously validated Kr antibodies (58 [↗](#), 59 [↗](#)), as well as a Kr::GFP(Bac) reporter line, which carries a bacterial artificial chromosome containing an approximately 88-kb region on the second chromosome (2R: 25168930 to 25256753) encompassing the entire Kr gene locus (2R:25226611 to 25231394), in which the Kr gene is fused to a GFP-FLAG-PreScission-TEV-BLRP tag sequence at its 3' end (60 [↗](#)). Indeed, this reporter construct successfully rescued the viability of a Kr null mutant (*Kr¹/Df(2R)Kr10*), confirming its functionality. We used *OK107>mCD8::GFP* to label MBNBs and traced Kr expression in these cells from embryogenesis to pupal stages. In mid-to-late embryogenesis, Kr was detected in thoracic neuroblasts, consistent with previous reports, confirming the specificity of our antibodies (stage 12 embryos, shown as a representative, Fig. S3a, b [↗](#)). However, in the dorsoanterior brain region, where MBNBs reside, Kr was absent from MBNBs, although it was occasionally detected in *OK107>GFP*-positive neurons, which lacked Mira (Fig. S3a, b [↗](#)). This suggests that while Kr is expressed in a subset of early-born neurons, it is not present in MBNBs during embryonic development.

We also examined Kr expression in postembryonic stages. In third-instar larval brains, Kr was weakly detected in MBNBs using both Kr antibodies (Fig. S3C [↗](#)). The Kr::GFP(Bac) reporter also exhibited stronger Kr::GFP signals in single cells adjacent to MBNBs, which likely represent GMCs or immature neurons that have inherited Kr::GFP from MBNBs (Fig. S3D [↗](#)). In addition to MBNBs, strong nuclear Kr signals were also specifically observed in a few neuronal cells attached to the mushroom body calyx (Fig. S3E [↗](#)). In pupal brains, similar weak Kr signals were detected in MBNBs, with a potential increase in expression around 24–48 hours after pupal formation (APF), as shown by both Kr antibodies and Kr::GFP(Bac) (Fig. 3A, B [↗](#)). To further validate these findings, we analysed publicly available gene expression datasets, including FlyAtlas2, modENCODE, and MBNB-lineage-specific RNA-seq data (30 [↗](#), 61 [↗](#), 62 [↗](#)). These datasets supported our observations, showing low but detectable levels of Kr expression in MBNBs during larval and pupal stages. These data indicate that Kr is likely absent in MBNBs during embryonic development but is expressed at low levels during larval and pupal stages. This pattern contrasts with that of canonical tTFs, which exhibit strong expression during neuroblast proliferation windows (25 [↗](#)), suggesting that Kr does not behave as a typical tTF in MBNBs.

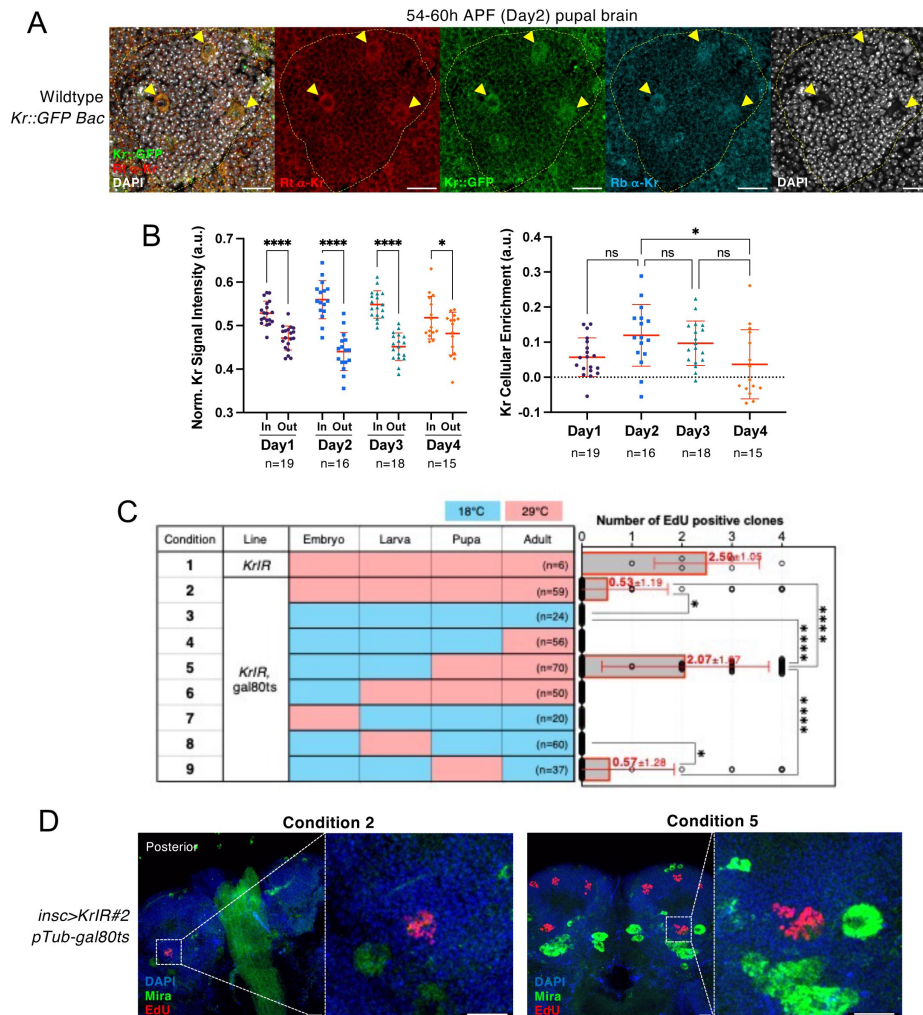


Figure 3.

Kr is expressed in MBNBs and regulates their elimination during the pupal stage

(A) Representative images of the MB cell body region in wild-type pupal brains (54–60 h APF) expressing the *Kr::GFP(Bac)* reporter and stained with two Kr-specific antibodies (rat anti-Kr: red, rabbit anti-Kr: cyan) and DAPI (grey). MBNBs, identified by their cell size and position within the MB cell body region, exhibit weak Kr signals detected by both antibodies and *Kr::GFP* (green). Scale bar: 20 µm. **(B)** Quantification of Kr expression in MBNBs during pupal development. (Left) Scatter dot plots showing normalised Kr signal intensities inside and outside individual MBNBs at different pupal stages. (Right) Kr signal enrichment in MBNBs, calculated as the ratio of intra-MBNB signal intensity to its surrounding area. The means and standard deviations are shown by thick and thin red bars, respectively. 'n' represents the number of brain hemispheres analysed per condition. Statistical analyses were performed using two-way ANOVA with Šidák's post-hoc test (left panel) and one-way ANOVA with Tukey's post-hoc test (right panel). Significance levels: * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p < 0.0001$, ns = non-significant. See Materials and methods for details on quantification. **(C)** (Left) Experimental design for the temporal knockdown of Kr during different developmental stages using *insc-Gal4*, alongside *pTub-Gal80ts*, and *Kr RNAi* (*KrIR#2*). A table summarises nine experimental conditions with temperature shifts: flies raised at 29°C (pale red boxes) induced *Kr RNAi*, while those at 19°C (pale blue boxes) suppressed RNAi by *Gal80ts*. 'n' indicates the number of brain hemispheres used for EdU-positive clone counting in each condition. (Right) Quantification of EdU-positive clones in adult brains from different experimental conditions. Scatter dot plots show the number of EdU-positive clones per hemisphere. Greyed boxes and red bars represent the means and standard deviations, respectively, with actual values annotated above. n' represents the number of hemispheres analysed per condition. Pair-wise Mann-Whitney U tests was used for statistical analysis. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p < 0.0001$. **(D)** Representative images of posterior adult brain regions from Group 2 (control) and Group 5 (*Kr RNAi* during pupal stage) conditions in (C), stained for DAPI (blue), Mira (green), and EdU (red). Persistent EdU-positive clones were observed in *Kr RNAi* conditions. Scale bar: 100 µm.

Next, we sought to determine whether Kr functions at a specific developmental stage to regulate MBNB elimination. Using the Gal4-Gal80ts system (63), we temporally controlled the knockdown of Kr by RNAi at different stages of development (Fig. 3C, D). We found that Kr RNAi induction specifically during the pupal stage was sufficient to induce MBNB retention in adult brains (Conditions 5 and 9 in Fig. 3C, D), while its knockdown at earlier embryonic or larval or later adult stages had no effect (Conditions 4, 7 and 8). This suggests that Kr plays a critical role during the pupal stage to ensure the timely cell cycle exit and subsequent elimination of MBNBs. Interestingly, continuous Kr RNAi induction from early larval stages suppressed MBNB retention (Conditions 2 and 6 in Fig. 3C, D), indicating that Kr may have an additional, potentially opposing function in NB regulation during larval stages.

Together, these results demonstrate that Kr functions during the pupal stage to promote MBNB elimination, a role that is distinct from its previously characterised functions. The low but persistent expression of Kr in MBNBs suggests that it acts in a unique lineage-specific manner, controlling their fate transition and elimination at the appropriate developmental time. The observation that pupal-stage Kr depletion is sufficient to induce MBNB retention, while early larval depletion can suppress this phenotype, highlights the complexity of Kr function in neuroblast regulation and suggests that its role extends beyond simple temporal patterning.

Kr Promotes MBNB Cell Cycle Exit and Elimination Through the Regulation of Imp

Having established that Kr function during the pupal stage is essential for MBNB elimination, we next sought to investigate the molecular mechanism underlying this regulation. Postembryonic NB elimination is governed by both cell-intrinsic and extrinsic factors, among which the RNA-binding proteins Imp and Syp play a particularly well-established role in NB survival, cell cycle exit, and neuronal fate specification. These two factors function in opposing temporal gradients, coordinating both NB persistence and neuronal fate transitions during development (64). In MBNBs, Imp is highly expressed during early postembryonic stages, where it promotes MBNB proliferation and biases neuronal progeny toward the early γ -neuronal fate. As development progresses, Imp levels decline, while Syp expression increases, facilitating the transition from $\alpha'\beta'$ to $\alpha\beta$ neuronal fate and promoting MBNB cell cycle exit and elimination (29,30). Given the established role of the Imp-Syp gradient in NB fate regulation, we hypothesised that Kr may interact with this pathway to control MBNB elimination.

We first examined Imp expression in *insc>KrIR* and *Kr^{Jf-1}* mutant adult brains. In both conditions, the Imp-expressing area in the MB cell body region was markedly expanded compared to controls. Notably, mitotic MBNBs were embedded within this expanded Imp-positive domain (Fig. 4A), suggesting a strong correlation between persistent MBNB activity and increased Imp expression. In contrast, the overall size of Syp-expressing cell populations appeared largely unchanged in Kr-depleted brains, although Syp expression levels were noticeably reduced in the MB cell body region in *insc>KrIR* brains compared to controls (Fig. S4A). To quantify the expansion of Imp-expressing domain, we measured Imp-expressing tissue area in the MB cell body region on the adult brain surface, using *OK107*-Gal4-driven GFP as a MB lineage marker. This analysis confirmed a statistically significant increase in Imp-positive area upon Kr depletion via *insc>KrIR* (Fig. 4B). Notably, in this experiment, *OK107*-Gal4 was primarily used to mark the MB lineage areas rather than to drive KrIR expression. When *KrIR* was driven by *OK107*-Gal4 itself, it did not induce MBNB retention or affect Imp expression patterns (*OK107>KrIR#2*, Fig. 4B). This is likely due to the significantly weaker induction ability of *OK107*-Gal4 in MBNBs compared to their differentiated progeny in postembryonic stages, as illustrated in Figs. 2F and S3C.

To determine whether Imp expression is maintained in the retained MBNBs in *insc>KrIR* and *Kr^{Jf-1}* adult brains, we directly examined Imp and Syp expression in these mitotic NBs. Strikingly, nearly all retained MBNBs continued to express Imp, while also co-expressing Syp (Figs. 4C-E). This is a

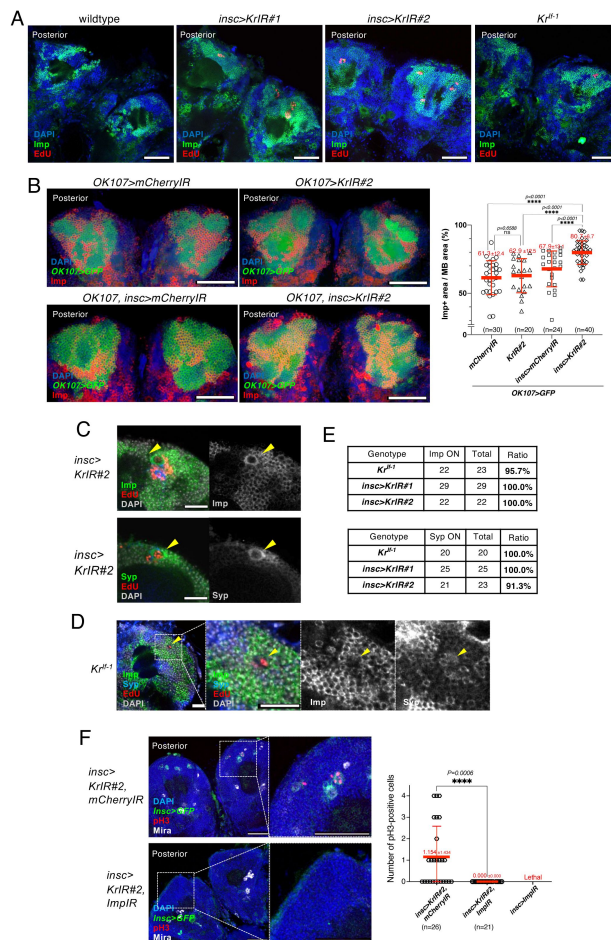


Figure 4.

Kr regulates Imp expression to promote MBNB proliferation and survival.

(A) Representative confocal images of dorsoposterior regions of wild type, *Kr^{IF-1}* mutant, and *insc>KrIR* adult brains. EdU labelling (red) marks proliferating cells, and Imp expression (green) was visualised using an Imp-specific antibody. DAPI (blue) stains nuclei. Compared to controls, Imp-expressing regions around the MB cell body region appear expanded in *Kr^{IF-1}* and *insc>KrIR* brains. Scale bars: 50 μ m. **(B)** Quantification of Imp-expressing areas within MB cell body regions. (Left) Representative 3D reconstructions of adult brain dorsoposterior surfaces for the indicated genotypes, with Imp (red), DAPI (blue), and OK107-Gal4-driven GFP (green) marking MB lineages. Scale bars: 100 μ m. (Right) Scatter dot plots showing Imp+ area ratios to MB cell body regions on the dorsoposterior brain surface, quantified per hemisphere. Thick and thin red bars indicate means and SDs, respectively. 'n' represents the number of hemispheres analysed per condition. Mann-Whitney tests were performed for statistical significance. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p < 0.0001$, ns = non-significant. **(C-E)** Imp and Syp expression in persisting MBNBs in *Kr* RNAi and *Kr^{IF-1}* mutant adult brains. **(C)** Representative image of active MBNBs in *insc>KrIR#2* brains, immunostained for Imp (upper panels) and Syp (lower panels). Merged images show Imp/Syp (green), EdU (red), and DAPI (grey), with individual Imp or Syp channels in grey. Arrowheads indicate mitotic MBNBs, identified by EdU signals and cell morphology. Scale bars: 20 μ m. **(D)** Representative confocal images of *Kr^{IF-1}* mutant adult brains, showing co-expression of Imp (green) and Syp (blue) in active (EdU+) MBNBs. Arrowheads indicate mitotic MBNBs. Single-channel Imp and Syp signals are shown in grey. Scale bars: 20 μ m. **(E)** Counting of mitotic MBNBs expressing Imp and/or Syp in *Kr^{IF-1}* and *insc>KrIR* adult brains. Upper and lower tables show the number of Imp+ and Syp+ mitotic MBNBs, respectively. Almost all retained MBNBs co-expressed Imp and Syp at high levels, an atypical phenotype for NB progression. **(F)** Imp depletion suppresses MBNB persistence in *Kr* RNAi brains. (Left) Representative confocal images of adult brains from *insc>KrIR#2, mCherryIR* (upper panels) and *insc>KrIR#2, ImpIR* (lower panels), immunostained for pH3 (red), *insc>GFP* (green), Mira (grey), and DAPI (blue). While mitotic MBNBs persisted in *insc>KrIR#2, mCherryIR* brains, they were absent in *insc>KrIR#2, ImpIR* brains. Scale bars: 100 μ m. (Right) Scatter dot plots of mitotic MBNB (pH3+) counts per hemisphere, with means and SDs depicted by thick and thin red bars, respectively. 'n' represents the number of hemispheres analysed. Mann-Whitney test: $p = 0.0006$ (****). *insc>ImpIR* alone was lethal, preventing adult brain analysis.

highly unusual phenotype, as Imp and Syp normally function in a mutually exclusive manner. However, previous studies suggest that Imp and Syp does not directly regulate each other, as over-expression of Imp or Syp does not repress the expression of the other (30). This raises the possibility that Kr may specifically regulate Imp, rather than directly influencing the Imp-Syp switch.

To determine whether persistent Imp expression is required for MBNB survival in Kr-depleted brains, we performed co-depletion experiments by expressing both *Kr*RNAi and *Imp*RNAi in NBs. While Imp depletion alone (*insc>ImpIR*) resulted in larval lethality, co-depletion of Kr and Imp (*insc>ImpIR, KrIR*) partially rescued this lethality, allowing the emergence of a small fraction of adult flies. However, these flies died within a few days of eclosion, preventing reliable EdU incorporation assays. To overcome this limitation, we assessed the presence of mitotic MBNBs using pH3 immunostaining in surviving young adult flies. Notably, while we observed mitotic MBNBs in *Kr*RNAi adult brains (*insc>KrIR#2, mCherryIR*), Imp co-depletion completely suppressed this phenotype (*insc>KrIR#2, ImpIR*, Fig. 4F, S4B). These results provide strong functional evidence that persistent Imp expression is essential for the maintenance of MBNBs in *insc>KrIR* adult brains.

Together, these findings demonstrate that Kr functions as a key regulator of MBNB elimination by promoting the downregulation of Imp, a prerequisite for proper Imp-Syp transition and neuroblast termination.

Kr Antagonises Kr-h1 to Promote MBNB Cell Cycle Exit

Having demonstrated that Kr depletion extends MBNB persistence through the dysregulation of Imp, we next sought to determine how Kr regulates Imp expression and MBNB fate. While Imp is widely expressed in various NBs, not just MBNBs (Liu et al., 2015), our data indicate that Kr specifically regulates cell cycle exit of MBNBs, we hypothesised that Kr may act through transcriptional regulation in a lineage-specific manner, possibly interacting with other transcription factors.

One promising candidate is Kr-h1, another KLF transcription factor that has been implicated in various aspects of neural development, including in the MB development (39,40). Kr-h1 is a well-characterised hormonal signalling target, particularly in response to juvenile hormone (JH) and ecdysone, where it counteracts the activity of E93, an ecdysone-induced transcription factor. E93, an ecdysone-induced transcription factor (37,65). Importantly, E93 has been shown to be required for timely elimination of MBNBs through autophagy regulation (32). MB lineage-specific RNAseq data indicate that Kr-h1 expression declines sharply after the larval-pupal transition (30), suggesting a possible involvement in regulating MBNB elimination. Additionally, Kr-h1 was identified in a genetic screen as an enhancer of the *Kr^{If-1}* eye phenotype, pointing to an antagonistic relationship between Kr and Kr-h1. Given these observations, we hypothesised that Kr-h1 opposes Kr function in MBNBs, promoting NB proliferation and self-renewal.

To test this hypothesis, we first examined whether Kr-h1 is required for normal MBNB elimination. Kr-h1 knockdown alone (*insc>Kr-h1IR*) did not prevent MBNB elimination, consistent with previous findings that Kr-h1 is largely dispensable for MB development (39,40) (Fig. 5A). However, when Kr-h1 RNAi was co-expressed with Kr RNAi (*insc>KrIR, Kr-h1IR*), the number of retained MBNBs in adult brains was significantly reduced compared to Kr knockdown alone, although they were not completely eliminated (Fig. 5A). This suggests that Kr-h1 contributes to MBNB persistence in the absence of Kr, underpinning an antagonistic interaction between Kr and Kr-h1 in MBNB cell cycle regulation.

To further examine Kr-h1's function, we overexpressed Kr-h1 (*Kr-h1::flag*) in NBs using *insc*-Gal4. Strikingly, Kr-h1 induction led to a dramatic increase in the number of proliferating NBs, including MBNBs, in the adult central brain, a phenotype markedly more severe than that observed in Kr

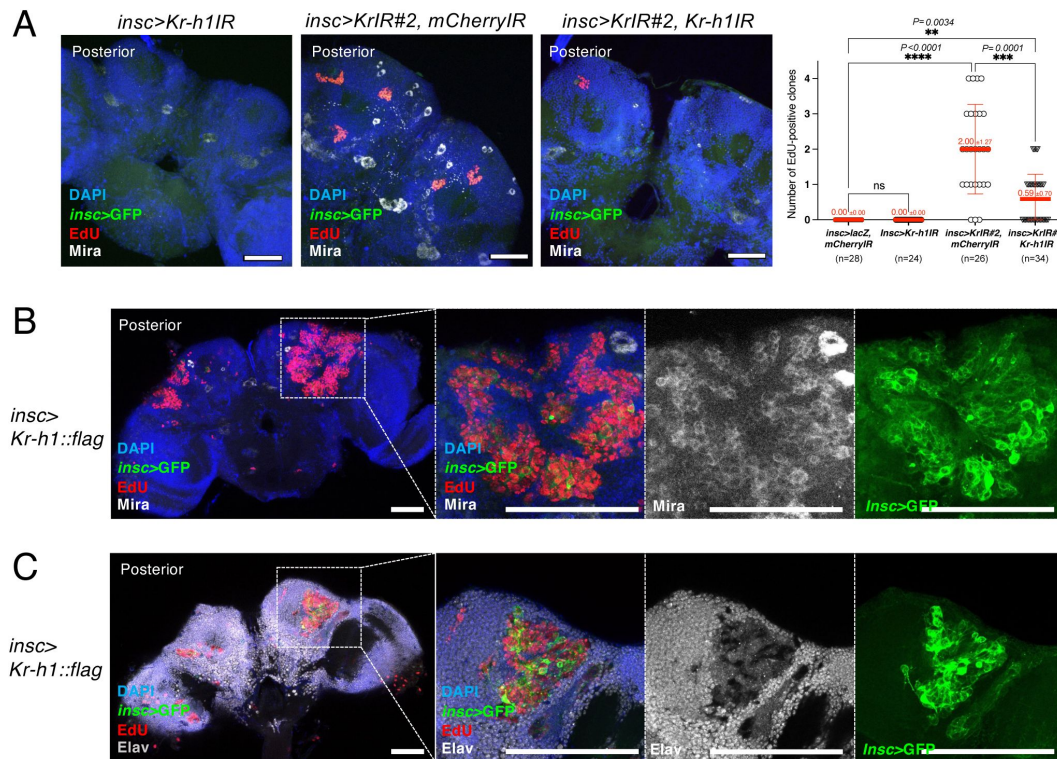


Figure 5

Kr antagonises Kr-h1 to promote MBNB cell cycle exit and prevent neuroblast overgrowth.

(A) Kr-h1 co-depletion suppresses MBNB persistence in *Kr* RNAi adult brains. (Left) Representative confocal images of *insc>Kr-h1IR*, *insc>KrIR#2, mCherryIR*, and *insc>KrIR#2, Kr-h1IR* adult brains, immunostained for EdU (red), *insc>GFP* (green), Mira (grey), and DAPI (blue). While *Kr-h1* RNAi alone did not affect MBNB elimination, the number of persisting EdU-labelled MBNB clones observed in *insc>KrIR#2, mCherryIR* brains was significantly reduced when *Kr-h1* was co-depleted in NBs. Scale bars: 100 μ m. (Right) Quantification of EdU-positive MBNB clones per hemisphere in indicated genotypes. Scatter dot plots display individual data points, with means and SDs represented by thick and thin red bars, respectively. 'n' represents the number of hemispheres analysed. Kruskal-Wallis test followed by Dunn's post-hoc test. **Significance levels: ** $p \leq 0.05$, *** $p \leq 0.01$, **** $p \leq 0.001$, **** $p < 0.0001$, ns = non-significant. **(B, C)** Kr-h1 overexpression promotes MBNB proliferation and blocks differentiation, leading to a neuroblastoma-like phenotype. **(B)** Representative posterior views of adult brains overexpressing Kr-h1 in NBs (*insc>Kr-h1::flag*), immunostained for DAPI (blue), *insc>GFP* (green), EdU (red), and Mira (grey), with magnified views of the MB cell body regions (right panels). Most EdU-positive clones were composed of cells co-expressing Mira and *insc>GFP*, indicating their NB-like status and a failure in differentiation. Scale bars: 100 μ m. **(C)** Kr-h1 overexpression inhibits neuronal differentiation. Confocal images of *insc>Kr-h1::flag* adult brains stained for DAPI (blue), *insc>GFP* (green), EdU (red), and Elav (grey). Unlike control brains, EdU-labelled clones in Kr-h1-overexpressing brains contained very few Elav-positive neurons, indicating a block in neuronal differentiation. Scale bars: 100 μ m.

depletion (**Fig. 5B, C**). These data indicate that Kr-h1 promotes NB proliferation and prevent their elimination.

Notably, beyond its effect on NB persistence, Kr-h1 overexpression also disrupted normal asymmetric division and differentiation of NBs. In *insc>Kr-h1::flag* adult brains, nearly all cells within EdU-labelled clones expressed Mira and *insc*-Gal4-driven GFP, while very few cells expressed Elav, suggesting a failure in neuronal differentiation (**Fig. 5B, C**). This phenotype resembles NB tumour-like overgrowths observed in *brat*, *numb* or *pros* mutant NBs, in which progeny fail to differentiate and instead retain a progenitor-like state (66).

Together, these findings suggest that Kr and Kr-h1 function antagonistically in MBNBs: Kr promotes MBNB cell cycle exit, while Kr-h1 promotes their proliferation and self-renewal. Kr-h1 may act upstream of, or in parallel to, the Imp-Syp axis, opposing Kr-mediated MBNB elimination.

Kr contributes to the MB development

Our findings indicate that Kr promotes MBNB cell cycle exit by repressing Imp and counteracting Kr-h1 during the pupal stage. Given that Imp also plays a key role in MB temporal identity transitions, we hypothesised that Kr may influence MB development beyond its role in NB elimination.

To test this, we examined MB morphology in *insc>KrIR* and *OK107>KrIR* adult brains. These flies developed normally, and *OK107>GFP*-labelled MB structures showed no significant abnormalities in their overall architecture (**Fig. 6A, Fig. 4B**). Despite this, when we visualised the α/β lobes using Fasciclin II (FasII), they appeared noticeably thinner and more curved in *insc>KrIR* adult brains compared to controls (**Fig. 6B**). Given that Imp promotes early neuronal fate, its persistent expression in MBNBs due to Kr depletion may interfere with the normal transition of MB neuronal identities, leading to defects or a delay in α/β lobe formation.

To further explore the role of Kr in MB development, we tested the effects of Kr overexpression. Overexpression of V5-fused Kr (Kr::V5) by *insc*-Gal4 resulted in early larval lethality, preventing further analysis. To circumvent this, we used *OK107*-Gal4, which has lower activity in MBNBs than *insc*-Gal4, to drive Kr overexpression specifically in MB lineages. Similar to *insc>Kr::V5*, *OK107>Kr::V5* also caused high lethality, though a small proportion of adult flies survived. These surviving flies lacked mitotic MBNBs, indicating that excessive Kr expression does not prevent MBNB elimination. Nevertheless, unlike the normal MB morphology observed in *OK107>KrIR#2* and *Kr^{If-1}* mutant adult brains, the brains of surviving *OK107>Kr::V5* adult flies exhibited severe morphological defects in the MB, with the α/β and α'/β' lobes significantly reduced and the γ lobes extensively disorganised (**Fig. 6C**). These results suggest that Kr not only promotes MBNB elimination but also plays a crucial role in ensuring proper neuronal fate transitions and MB structure formation, partly mediated by its regulation of Imp.

Discussion

Kr Functions in the Timely Elimination of MBNBs to Prevent Adult Neurogenesis

Our study reveals a previously unrecognised role for Kr in orchestrating the timely elimination of MBNBs during *Drosophila* development, thereby preventing these progenitors from persisting and proliferating into adulthood. In contrast to its well-characterised function as a gap gene and early tTF in embryonic NBs, we show that Kr functions at low levels during the pupal stage to repress Imp expression, facilitate MBNB cell cycle exit, and drive MBNB elimination (**Fig. 6D**). Furthermore, the classic *Irregular facet* (*Kr^{If-1}*) mutation, which misregulates Kr expression

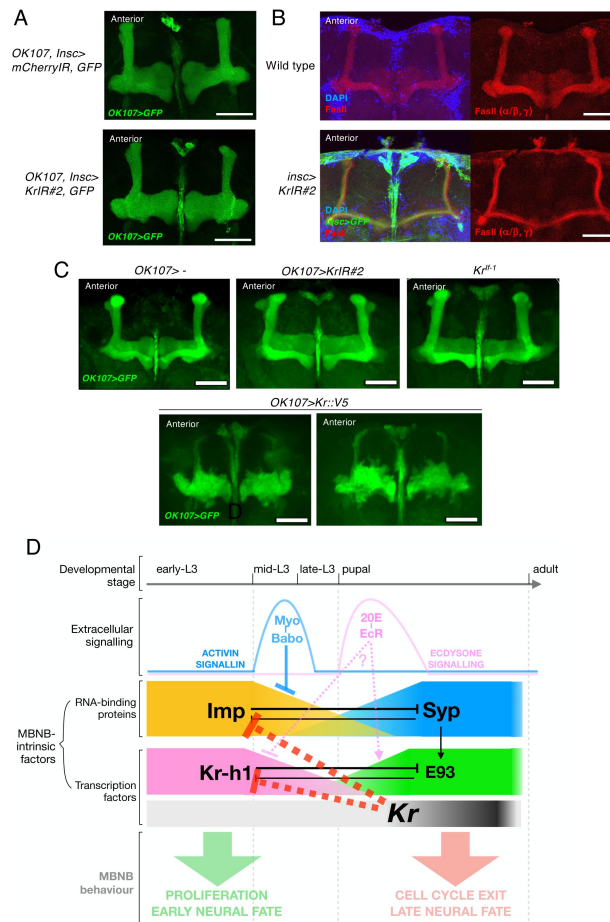


Figure 6.

Kr regulates MB development by promoting MBNB elimination and facilitating neuronal fate transitions.

(A) Representative confocal images of anterior views of adult brains from *OK107, insc>mCherryIR, GFP* (control) and *OK107, insc>KrIR#2, GFP* flies, with MB lobe structures visualised by *OK107>GFP* (green). No overt structural abnormalities were observed in MB lobes upon Kr depletion. Scale bars: 50 μ m. **(B)** Representative images of MB α/β lobe morphology in wild-type and *insc>KrIR#2, GFP* adult brains, visualised by Fasciclin II (FasII) staining (red). Kr depletion led to thinner and more curved α/β lobes compared to controls. Scale bars: 50 μ m. **(C)** Effects of Kr overexpression in MB lineages on MB development. Anterior views of 3D-reconstructed confocal images of adult brains from control (no UAS gene), *OK107>KrIR#2, Kr^{lf-1}* mutant and *OK107>Kr::V5* flies, with MBs visualised by *OK107>GFP* (green). While control, Kr knockdown and *Kr^{lf-1}* mutant adult brains exhibited normal MB morphology, Kr overexpression (*OK107>Kr::V5*) led to high lethality. In the few surviving adult flies, there was severe disorganisation of MB lobes, particularly with pronounced reduction in α/β and α'/β' lobes and extensive disarray in γ lobes. Scale bars: 50 μ m. **(D)** A proposed model for Kr function in MBNB regulation. During early developmental stages, specifically from mid-L3 to late L3, elevated levels of Imp and Kr-h1 are key drivers of MBNB proliferation and early neuronal fate specification. As development transitions into the pupal stage, Kr assumes a critical role by repressing Imp and counteracting Kr-h1. This dual action by Kr facilitates cell cycle exit and ensures proper transitions in MB neuronal fate. Further complexity in MBNB regulation is introduced by extracellular signaling pathways. The Activin/Baboon pathway contributes to the suppression of Imp, enhancing the transition control exerted by Kr. Concurrently, delayed ecdysone signaling, mediated through the Kr-h1-E93 pathway, plays a pivotal role in facilitating MBNB elimination. Kr, functioning as a transcription factor, may endow MBNBs with the specific competence necessary to integrate and coordinate these extracellular signals, which are essential for regulating MBNB proliferation and fate transitions. The absence of Kr disrupts this finely tuned regulatory mechanism, leading to continued MBNB proliferation and neurogenesis in adult brains. This disruption may also delay the transition of MB neuronal fates, underscoring Kr's integral role in maintaining developmental timelines and structural integrity within the MBs.

(41 [↗](#), 42 [↗](#)), leads to sustained MBNB maintenance and neurogenesis in adult brains. This represents, to our knowledge, the first spontaneous mutation identified in metazoans that alters adult neurogenesis. These findings highlight MBNBs as a uniquely flexible neurogenic population within the *Drosophila* brain and demonstrate how transcriptional misregulation can unlock latent neurogenic potential.

The Adult *Drosophila* Central Brain Is Predominantly Postmitotic but Retains Latent Neurogenic Capacity

Consistent with previous studies (9 [↗](#), 16 [↗](#), 17 [↗](#)), our analysis confirms that the adult *Drosophila* central brain is largely postmitotic, with no detectable mitotic figures or EdU incorporation in control adult brains (Fig. 1 [↗](#)). Nearly all cells display G1-phase markers, reflecting strict control over cell cycle re-entry. However, occasional cells exhibit G2-phase markers in the absence of NB markers, suggesting a population of quiescent progenitors that may be reactivated under specific conditions, such as injury (43 [↗](#)–45 [↗](#)).

To further explore the basis of this postmitotic state, we analysed CCR expression using scRNA-seq data (48 [↗](#), 49 [↗](#)). Our findings reveal widespread suppression of pro-proliferative regulators, including cyclins and Cdks, in adult neuronal and glial clusters, alongside increased expression of inhibitors such as *Wee1* and *fzr* (Fig. 1 [↗](#), S1 [↗](#)). This coordinated repression likely prevents unintended proliferation and preserves neural integrity. Indeed, forcing the expression of *E2F1-Dp* and *Cdk2-CycE* drove mature neurons into mitosis (Fig. S2 [↗](#)), but at a high cost—many cells underwent apoptosis, and flies exhibited reduced lifespan. These findings suggest that, although adult neurons retain some intrinsic proliferative potential, overriding postmitotic regulation leads to deleterious consequences.

A Distinct Late Role for Kr in MBNB Termination

A key finding of this study is that Kr depletion results in MBNBs persisting into adulthood, as evidenced by mitotic figures and EdU-labeled progeny co-expressing the NB marker Miranda. These retained MBNBs continue asymmetric division and generate Elav-positive neurons that integrate into the MB (Fig. 2 [↗](#)). This contrasts sharply with normal development, where MBNBs are eliminated by late pupal stages (9 [↗](#), 16 [↗](#)). In stark contrast, *Kr* RNAi depletion or the *Kr^{If-1}* allele results in a complete bypass of MBNB elimination rather than a mere delay, underscoring Kr's essential role in terminating MB neurogenesis.

Although Kr has been widely studied as an early tTF in embryonic NBs, our findings indicate that it fulfills a distinct pupal-stage-specific role in MBNBs (Fig. 3 [↗](#)). Notably, Kr expression is undetectable in embryonic MBNBs (Fig. S3 [↗](#)), suggesting that Kr does not act as a conventional tTF in this lineage. Instead, it governs their transition toward elimination in conjunction with other regulatory cues such as ecdysone signaling (27 [↗](#), 28 [↗](#)). This aligns with the notion that transcriptional regulators can be reused or repurposed in discrete developmental windows to control different processes.

Kr as a Central Regulator of MBNB Termination through Imp Repression and Kr-h1 Antagonism

Mechanistically, our results place Kr as a crucial suppressor of Imp in pupal MBNBs (Fig. 6D [↗](#)). The opposing Imp-Syp gradient is fundamental to NB fate decisions: Imp promotes early NB proliferation and younger neuronal identities, while Syp facilitates late neuronal fates and NB termination (29 [↗](#), 30 [↗](#)). MBNBs are unique among NBs in maintaining proliferative capacity until late pupal stages. In Kr-depleted brains, MBNBs continue to express Imp, failing to transition toward termination. Co-depleting Imp alongside Kr rescues the MBNB-retention phenotype (Fig. 4 [↗](#)), confirming that persistent Imp is necessary for sustaining NB proliferation in the absence of Kr.

Beyond *Imp* repression, we uncover an antagonistic interplay between Kr and another KLF member Kr-h1 in regulating MBNB proliferation (Fig. 5, 6D). Overexpressing Kr-h1 induces excessive NB-like overproliferation, with tumor-like formations of progenitor-like cells lacking proper neuronal differentiation. Conversely, knocking down Kr-h1 partially rescues the MBNB-retention phenotype in Kr-depleted brains. Kr-h1, strongly modulated by juvenile hormone, can oppose ecdysone-induced transcription factors such as E93 (37,38). Notably, Kr-h1 appears dispensable for overall MB development (39), its removal can rescue TGF- β /Babo loss-of-function phenotypes that compromise MB neuron fate transitions (40), suggesting a close functional tie between Kr-h1 and Activin/Babo signaling. Indeed, Babo regulates timely *Imp* downregulation in MBNBs (31); thus, *Babo* or *Kr* depletion leads to a similar outcome—prolonged *Imp* expression and disrupted α/β fate specification. The partial rescue of both Babo and Kr knockdown phenotypes by Kr-h1 depletion further supports the idea that Kr mediates or cooperates with Activin/Babo signaling in MBNBs. Moreover, because E93 drives MBNB clearance through autophagy (32), Kr-h1 may extend NB persistence by suppressing E93. Future studies should clarify whether the Kr/Kr-h1/E93 axis interacts with additional systemic signals to coordinate NB lifespan and lineage progression.

Notably, MBNBs exhibit distinct proliferation patterns, including uninterrupted proliferation during the embryo-to-larval transition and their significantly delayed cell cycle exit compared to other NBs. In contrast to the typical ecdysone-driven *Imp* repression observed in most NBs, often mediated by EcR activated by Svp, the late postembryonic component of tTF series (27,28), MBNBs appear less responsive to early ecdysone cues. Instead, extracellular Activin/Babo signalling and E93, induced upon late ecdysone signalling, converge to regulate the ultimate *Imp*-to-Syp transition and NB termination, respectively (31,32). Our data suggest that Kr, expressed at low but relatively uniform levels in pupal MBNBs (Fig. 3A, B), acts as a permissive epigenetic factor, priming MBNBs to respond to Activin/Babo while dampening or delaying ecdysone responsiveness (Fig. 6D). Such a model would explain why Kr depletion both prolongs MBNB survival and disrupts proper *Imp* downregulation.

Further studies should explore whether Kr interacts with chromatin remodelers or transcription factor complexes in MBNBs, and how Kr-h1 might counteract Kr's permissive function at the epigenetic level. Dissecting the interplay of intrinsic (lineage-specific regulators) and extrinsic (hormonal) signals will be key to understanding how this unique MBNB regulatory circuit controls extended NB proliferation and fate specification.

Kr as a Multifaceted Regulator of MB Development

Beyond MBNB elimination, our data suggest that Kr influences MB development (Fig. 6). Subtle morphological defects arise when Kr is knocked down, while excess Kr disrupts MB lobe architecture, suggesting that tightly controlled Kr levels are critical for maintaining a balance between neuronal identity transitions and MB assembly. These findings imply that Kr not only ensures timely NB exit but also influences how MB neurons mature into distinct sublineages.

A key question is whether newly generated MB neurons in Kr-depleted or *Kr^{lf-1}* mutant brains adopt appropriate late α/β fates or experience delayed fate transitions. Since the *Imp*-Syp gradient governs neuronal fate progression, persistent *Imp* could bias these neurons toward an earlier identity or slow their transition to late identities. Given MBs' roles in olfactory learning and memory (13,14), sustained MB neurogenesis in Kr-depleted flies may alter memory formation or slow age-related MB decline. Future behavioral assays will be needed to determine whether extended MB neurogenesis has functional consequences for learning, plasticity, or ageing.

KLFs in Adult Neurogenesis and Plasticity

By demonstrating that Kr regulates MBNB elimination through Imp repression, Kr-h1 antagonism, and potential epigenetic priming of MBNBs for late-stage fate transitions, our study uncovers a novel function for Kr beyond its classical role as an early tTF. These findings contribute to the growing understanding that the adult brain, while largely postmitotic, retains latent neurogenic potential that can be reactivated under specific conditions or perturbations.

Beyond *Drosophila*, KLF family members are evolutionarily conserved. In mammals, KLF4 is a well-known pluripotency factor (67 [↗](#)) that protects NSCs from senescence (68 [↗](#)), while KLF9 maintains NSC quiescence and can non-cell-autonomously activate NSCs in the hippocampus (69 [↗](#), 70 [↗](#)). These parallels suggest that KLF proteins serve a broader role in calibrating progenitor maintenance and termination across species.

Understanding how the adult brain's default postmitotic state can be selectively overridden may offer new strategies for addressing neurodegeneration or brain injury. Future experiments exploring whether sustained MB neurogenesis in Kr-depleted flies enhances cognitive function or affects aging could provide insight into whether manipulated neurogenesis has adaptive or detrimental consequences.

Materials and methods

Fly stocks and culture

The *Drosophila* lines used in this study are as follows. Wild type and control stocks, *Oregon R* and *w¹¹¹⁸*, and a *Kr^{Jf-1}* mutant stock, *Kr^{Jf-1}/CyO*; *Sb/TM6B* were obtained from Cambridge Drosophila Facility. An additional *Kr^{Jf-1}* mutant stock *w¹¹¹⁸*; *Kr^{Jf-1}/CyO* (BCF360, from Shanghai Fly Center. *Kr* RNAi lines (v104150, v40871) from VDRC. The Gal4 driver lines specific for the mushroom body neurons, *mb247-Gal4* (a gift from Dr Margaret Ho) and *OK107-Gal4*; *UAS-mCD8::GFP* (a gift from Dr Yi Zhong). The neuroblast-specific Gal4 driver line *UAS-dicer2*; *insc-Gal4*, *UAS-mCD8::GFP/CyO* (a gift from Dr Eugen Knoblich). *pTub-gal80ts* lines (BL7107, BL7108), a GFP reporter line, *UAS-mCD8::GFP* (BL5130), a Kr overexpression line *UAS-Kr-V5* (BL83301) (Refs), a loss-of-function *Kr* mutant *Kr⁻¹* (BL3494) and a *Kr* deficiency line, *Df(2R)Kr10* (BL4961), and a *Kr::GFP(Bac)* reporter line *PBac[Kr-GFP.FPTB]* (BL56152) from the Bloomington Drosophila Stock Center. An *Imp* RNAi line (THU55645) from TsingHua Fly Center. *Kr-h1* RNAi line (BL50685) and *UAS-Kr-h1::FLAG* (kind gifts from Qianyu He). A Fly-FUCCI line *pUbp-GFP-E2F1*, *mRFP-NLS-CycB* (a kind gift from Dr Bruce Edgar).

All *Drosophila* lines were maintained and cultured with common cornmeal agar media, with 12 hr-12 hr day-night cycles, and at 50-70% humidity. All fly experiments were conducted at 25 °C except for RNAi and over-expression experiments in which flies were grown at 29°C for higher Gal4-dependent induction, as specifically noted. 20 to 30 flies were contained in each tube and flipped them into new tubes every two to three days. Fly crossing was done with around 10 virgins and 8 males with specific genotypes in each tube. In most crosses, virgins of Gal4 driver lines were collected and crossed with the male flies with UAS elements. The progeny with the specific genotypes were selected based on visible genetic markers for analysis.

Antibodies

Primary antibodies used in this study are as follows: rabbit Phospho-Histone H3 (Ser10) antibody (Invitrogen PA5-17869, 1:400), rat anti-Elav (Developmental Studies Hybridoma Bank, 7E8A10 1:50), rat anti-Mira (1:400) and rat anti-Dpn (1:100, kind gifts from Dr Chris Doe), rabbit anti-Kr antibody (a gift from Dr Chris Rushlow, 1:1000)(58 [↗](#)), rat anti-Kr antibody (from Asian

Distribution Center for Segmentation Antibodies, 1:100)([59](#), [71](#)), guinea pig anti-Syp (a gift from Ilan Davis, 1:200), rat anti-Imp and rabbit anti-Syp (kind gifts from Dr Claude Desplan, 1:200 and 1:1000), mouse anti-FasII (DSHB, 1D4, 1:40), rabbit anti-Cleaved *Drosophila* DCP-1 (Cell Signaling Technology #9578, 1:100), mouse anti-V5 antibody (Invitrogen, SV5-Pk1). Second antibodies used in this study are goat anti-mouse, rat or rabbit secondary antibodies conjugated with Alexa Fluor 488, 568 or 647 (Invitrogen, 1:500), DAPI (Cell Signaling Technology, 1:1000).

Immunofluorescence and confocal imaging of *Drosophila* brains and embryos

Drosophila adult brains were dissected in 1xPBS following the protocol described in Wu and Luo, 2006 ([72](#)). The brains were then fixed in freshly made fixative containing 8% formaldehyde, 1xPBS, 0.5mM EGTA and 5mM MgCl₂, at room temperature for 1 hour. The fixed brain samples were then washed in PBST (1xPBS, 0.3% Triton-X), blocked with 3% BSA in PBST (PBSTB) at room temperature for 1 hour, and then incubated in PBSTB containing appropriate primary antibodies at 4°C overnight. After washes, the samples were further incubated in PBSTB containing secondary antibodies and DAPI (Cell Signaling Technology) at room temperature for 3 hours, then washed in PBST and kept in mounting media (Vectashield, VectorLab). The brain samples were then mounted onto microscope slides and kept in 4°C till imaging.

As for pupal brains, the whole body of pupae with its shell removed were fixed in freshly made fixative containing 8% formaldehyde, at room temperature for 40 minutes, and after washing fixed pupal brains were then carefully dissected in 1x PBS.

Larval brain-eye complexes were fixed in freshly made fixative containing 4% formaldehyde, 1xPBS, 0.5mM EGTA and 5mM MgCl₂, at room temperature for 30 minutes. After washing, blocking, and staining steps mentioned as above, larval brains were been carefully separated from the complexes and mounted onto slides.

For embryos, parental flies were incubated in a small cage with an apple juice agar plate pasted with full of yeast extract paste, and replaced the old plate with a new plate with yeast every day. Collecting eggs at the desired stage of development by brushes and transfer them to a small basket. Dechorionate eggs by putting them in 50% bleach and wash the dechorionated eggs by water. The dechorionated embryos were then transferred into a glass bottle (1ml/6ml) containing 8%Flormaldehyde/PBS: Heptane (1:1) and incubated at room temperature for 20min. Incubating embryos 1x volume of -20°C MeOH for 1min after removing the lower FA phase. Washed the embryos with 1ml MeOH for 5min 3 times. The embryos were then rehydrated by 15min PBS and PBST (PBS +0.05%TritonX-100) incubation respectively. Immunostaining steps were the same as mentioned above, except embryos were blocked in PBTA(PBS +0.05%Triton X-100 + 1%BSA) at room temperature for 1 hr.

Images were taken by a Nikon C2 confocal microscope in four channels with emission laser wavelengths, 405, 488, 568 and 647 nm. Image reconstruction and analysis were processed and conducted using NIS-Elements AR 5.20 software and Image J. All projected images shown in this paper were processed by maximum intensity projection. For better visualisation, a 0.5 Gaussian blur filter was applied to every image shown.

EdU labelling of proliferating cells in *Drosophila* adult brains

EdU labelling was performed to detect proliferating cells in *Drosophila* adult brains, following the protocol outlined by Siegrist et al. (2010) ([16](#)). EdU powder was dissolved in a 5% sucrose solution to achieve a final concentration of 0.5 mg/ml, and this solution was applied to paper to replace the regular food source in the culture vials. One- to three-day-old young adult flies were then collected and raised until the flies reached certain ages, and then fed with EdU-supplemented

food for a period of three days prior to brain dissection. For EdU labelling chase experiments, EdU exposure was ceased after the initial labelling period, and the flies were subsequently maintained for an additional three, seven, or fourteen days before brain dissection.

Following dissection, adult brains were subjected to immunofluorescence staining using standard procedures, with the exception of an additional step for EdU detection. EdU Click labelling was performed in the dark for 30 minutes prior to primary antibody incubation, following the manufacturer's instructions (APEX-BIO EdU Imaging Kits Cy3, #K1075).

EdU-incorporated cell clones were visualised and manually counted in adult brains using a Nikon Ti2 fluorescence microscope equipped with a 20x objective lens. Statistical analyses were conducted using GraphPad Prism software. Data distribution was first assessed with the D'Agostino-Pearson omnibus normality test. For data that followed a normal distribution, an unpaired t-test was employed. For non-normally distributed data, the Mann-Whitney U test was utilized. Differences were considered statistically significant at $p < 0.05$. Significance levels were denoted as follows: * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p < 0.0001$; ns indicates non-significant results.

Confocal Imaging and Fluorescence Quantification of Kr Signal in MBNBs in *Drosophila* Pupal Brains (Fig. 3B [↗](#))

Wild type *Drosophila* lines carrying *Kr::GFP(Bac)* were maintained at 25°C. Pupae were collected over a 24-hour period and subsequently incubated for 0 (Day 1), 1 (Day 2), 2 (Day 3), or 3 days (Day 4) before the brains were dissected for immunostaining. The pupal brains were immunostained using rat anti-Kr (Alexa 568) and rabbit anti-Kr (Alexa 647) antibodies. Z-stack images encompassing the entire pupal brain were acquired from the posterior side using a Nikon C2 laser scanning confocal microscope with a 40× oil-immersion objective. MBNBs were identified based on their stereotactic localization on the dorsoposterior brain surface, distinct axonal projections of progeny cells visualized via DAPI counterstaining extending into the developing mushroom body calyces, and the presence of two symmetrical, rounded regions with low DAPI signal on the posterior brain, bordered by adjacent cell clusters exhibiting slightly elevated DAPI intensity.

The fluorescence intensity of the Kr signal (Alexa 568) in MBNBs was quantified using ImageJ software. DAPI channel images were used to define three regions of interest (ROIs) for each neuroblast: the nucleus, characterized by a central region with moderate DAPI signal; the cytoplasm, identified as the peri-nuclear area with negligible DAPI signal; and the extracellular region, noted as the background area adjacent to the neuroblast. Three independent measurements were taken for both cytoplasmic and extracellular regions of each MBNB, and the mean fluorescence intensity was calculated for each ROI category. Fluorescence intensity ratios for each MBNB were calculated and expressed using the following formulas: Normalized Kr (In) signal intensity = Cytoplasm / (Cytoplasm + Outside); Normalized Kr (Out) signal intensity = Outside / (Cytoplasm + Outside); Kr Cellular Enrichment = (Cytoplasm - Outside) / (Cytoplasm + Outside). Data are presented as mean $\pm$ SD. Statistical significance was determined using one-way ANOVA with Tukey's post-hoc test, performed in GraphPad Prism (v10.1.2). Source data files include pupal age (days after puparium formation), the number of MBNBs analyzed per group, and spatial coordinates of nine randomized measurement points per MBNB.

Quantification of Imp-Expressing Areas Within MB Cell Body Regions in *Drosophila* Adult Brains (Fig. 4B [↗](#))

Adult brains were dissected and fixed using a standard immunostaining protocol as described above. Brains from the *OK107-Gal4>UAS-mCD8::GFP* line were used to visualise MBNBs. The brains were immunostained using the rat anti-Imp antibody, with Alexa 568-conjugated secondary antibodies employed to visualise the Imp signal. GFP signal was detected natively from the

expressed *UAS-mCD8::GFP* construct. Brains were mounted with the posterior side facing the cover slip and imaged on a Nikon C2 confocal microscope. z-stacks were captured across the depth of the brain that included the entire MB cell body area and part of the MB calyx using a 1 μm z-step in three channels: DAPI, GFP, and Alexa 568 (Imp). 3D reconstructions were generated using Nikon NIS-Elements software with the alpha blending rendering function to visualise Imp and GFP signals on the brain surfaces. Images were adjusted to present the MB cell body region facing the viewer for consistent orientation and analysis. Imp and GFP signal intensities were thresholded to create clear binary images for subsequent quantification. The area of the GFP-positive, MB cell body regions was defined and measured using the Area function in NIS-Elements. Within these defined regions, the area exhibiting Imp positivity was also measured. The ratio of Imp-positive area to the total GFP-positive MB cell body area was calculated for each hemisphere, as depicted in scatter dot plots. The number of hemispheres analysed per condition is indicated in [Fig. 4B](#). Ratios of Imp-positive areas were compared using the Mann-Whitney test to assess statistical significance between experimental conditions. No preprocessing of data was necessary before statistical testing, as the data represented direct area ratios.

Differential Expression Analysis of Cell Cycle Genes in the *Drosophila* Adult and Larval Brains ([Fig. 1B](#), [S1](#))

The original data files of the single-cell RNA sequencing of *Drosophila* adult and larval brains ([48](#)) were obtained from the NCBI Gene Expression Omnibus (series number GSE107451) and processed using Seurat v3 ([73](#)). To ensure data quality and remove potential dying cells, we restricted the dataset to samples with 200 to 3700 unique genes and a UMI-to-gene ratio smaller than 6. The resulting dataset comprised 56,619 samples and 12,925 features. For normalisation, raw counts were first adjusted to account for sequencing depth by scaling each cell by the total number of counts and multiplying by a scaling factor of 10,000. The normalised counts were then log-transformed using the natural logarithm ($\log_1 p$), which transforms the data using the formula $\log(1 + x)$, where x represents the normalized counts. Principal Component Analysis (PCA) was conducted on the 2000 highly variable genes identified. Shared Nearest Neighbour (SNN) cell clustering at a resolution of 0.2 or 8 was performed using the first 20 principal components, and t-SNE reduction was applied to visualise the clustering results. Based on the differential expression of the cluster markers and comparison with the original annotations ([48](#)), 17 or 84 cell clusters were identified, respectively. The expression of 112 CCR genes across these clusters was then visualised using dot plots. In these plots, the colour intensity of each dot represents the average expression level of the CCR gene, with darker red indicating higher expression levels. The size of each dot indicates the proportion of cells within the cluster expressing the gene, with larger dots representing higher proportions. To illustrate the heterogeneity within clusters, heatmap plots were generated for each CCR gene. In these heatmaps, each gene is represented by a row and each cell by a short line, highlighting the variability of expression levels among individual cells within clusters.

For the analysis of *Drosophila* larval brains, the original data files of the single-cell RNA sequencing of larval brains ([49](#)) were obtained, processed and presented similar to the adult brain. The analysis dataset contains 4349 samples and 12,942 features. A resolution of 2 was used and 29 clusters were identified.

Survival Analysis of Flies Overexpressing Cell Cycle Regulators in Dopaminergic Neurons ([Fig. S2](#))

Survival were analysed following the overexpression of Dp-E2F1 and CycE-Cdk2 in dopaminergic neurons using the *TH-Gal4* driver combined with a Gal80ts system to achieve temporal control of gene expression. Experimental flies carried the genotype *TH-Gal4*, *UAS-Dp-E2F1*, *UAS-CycE-Cdk2*, *UAS-mCD8::GFP*; *tub-gal80ts*, allowing for targeted overexpression in dopaminergic neurons. Control flies expressed *lacZ* under the same driver system (*TH-Gal4*, *UAS-lacZ*, *UAS-mCD8::GFP*;

tub-gal80ts). Gene expression was induced by shifting the environmental temperature from 19°C to 29°C. This temperature shift was applied to adult flies aged 2 days post-eclosion to activate the Gal4, thereby initiating overexpression of the CCRs in the targeted neuron population.

Survival was monitored daily, and Kaplan-Meier survival curves were generated to compare the lifespans between the experimental and control groups. The experiment was conducted in three independent replicates to ensure data robustness. The log-rank (Mantel-Cox) test was used to determine statistical significance between survival distributions of the experimental and control groups. Error bars on the survival curves represent standard deviations, emphasising the variability and reliability of the observed effects.

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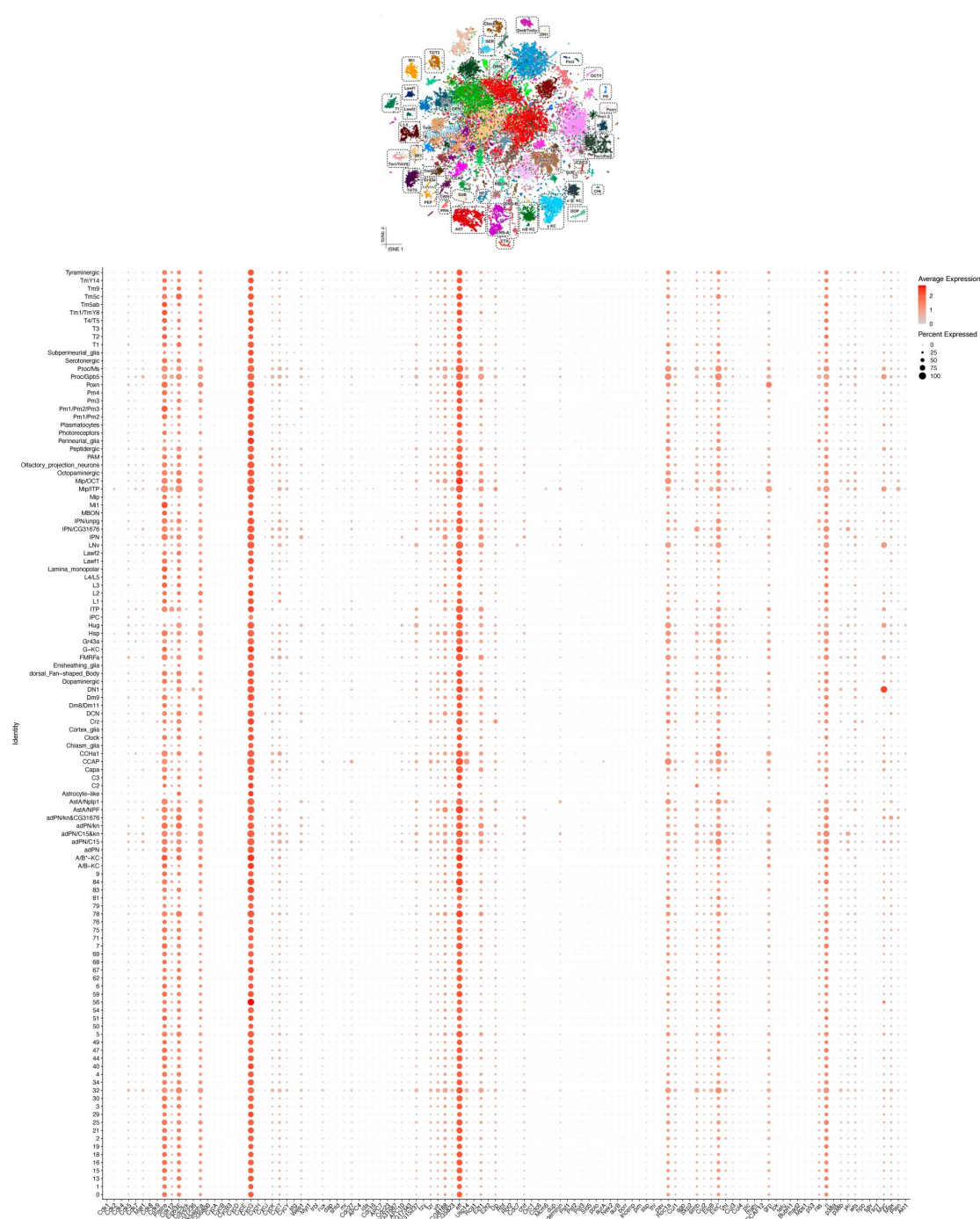


Figure S1.

Expression of CCR genes across all 84 cell clusters in the adult *Drosophila* brain.

(Top) tSNE plot depicting 84 cell clusters identified in the adult *Drosophila* brain (Davie et al., 2018). (Bottom) A dot plot illustrating the expression levels of 112 CCRs across 84 cell clusters. Each dot represents the average expression level of a CCR gene within a cluster, with colour intensity indicating expression level (darker red corresponds to higher expression), and dot size representing the proportion of cells within the cluster expressing the gene. The data show that positive regulators of the cell cycle, including *Cdk1*, *Cdk2*, *CycB*, and *Polo*, are largely absent from neuronal and glial clusters but are enriched in the small population of neuroblast-like cells. In contrast, negative regulators, such as *fzr*, *Wee1*, and *rux*, remain broadly expressed across neuronal clusters, reinforcing the molecular mechanisms that maintain the postmitotic state of the adult brain.

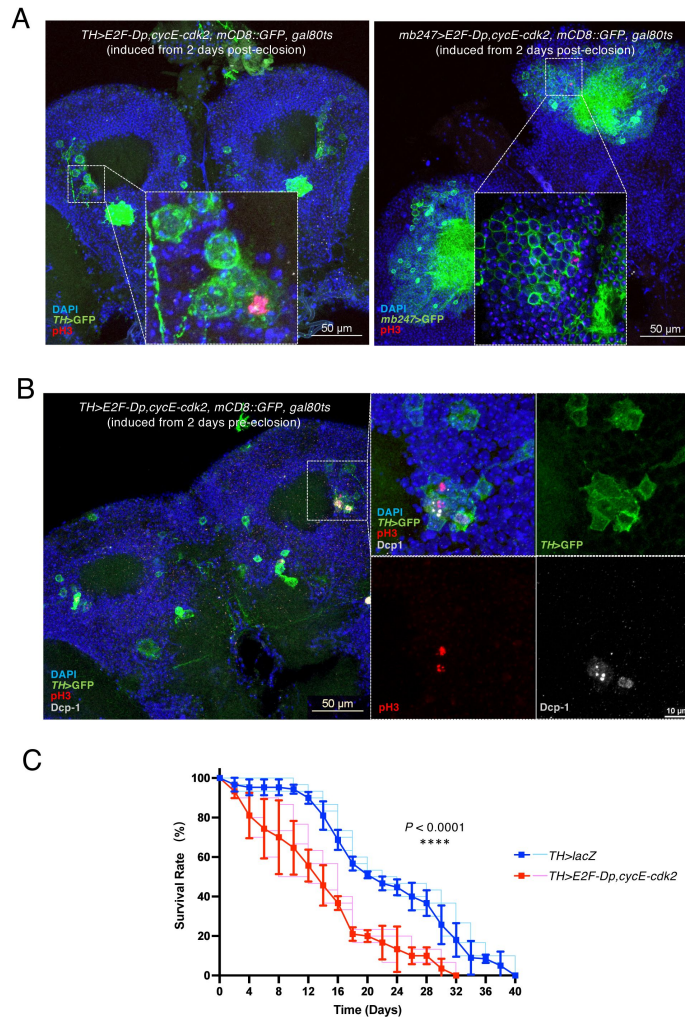


Figure S2.

Forced expression of positive CCRs induces mitotic entry in mature neurons.

(A) Overexpression of the positive CCR combination Dp-E2F1 and CycE-Cdk2 in dopaminergic neurons using *TH*-Gal4 (left panels) and in mushroom body neurons using *mb247*-Gal4 (right panels) induces mitotic entry in young adult flies (~2 days post-eclosion at 19°C). Flies were incubated at 29°C for 10 days to sustain CCR gene expression. pH3 immunostaining reveals mitotic neurons in these brains, whereas no pH3-positive neurons were observed in controls. However, when gene induction was performed in older flies (10 days post-eclosion at 29°C), no pH3-positive neurons were detected ($n \geq 10$), suggesting that neurons become increasingly refractory to cell cycle entry with age. Scale bars: 50 μ m. **(B)** Overexpression of Dp-E2F1 and CycE-Cdk2 in dopaminergic neurons during the late pupal stage (~2 days before eclosion) allowed some adult flies to emerge; however, these flies exhibited premature lethality, dying within 3–5 days post-eclosion. Immunostaining of dissected brains 5 days after gene induction revealed pH3-positive neurons, with nearly half of the examined flies showing positive signals (12 out of 27 brains examined). Some pH3-positive neurons also exhibited apoptotic marker Dcp-1 (grey), indicating that forced mitotic re-entry may induce cell death. Overexpression of the same CCR combination in the mushroom body using *mb247*-Gal4 resulted in 100% lethality before eclosion, preventing adult emergence. **(C)** Survival analysis of flies overexpressing positive CCRs in dopaminergic neurons. Kaplan-Meier survival curves comparing adult flies overexpressing Dp-E2F1 and CycE-Cdk2 in dopaminergic neurons (*TH>E2F-Dp, CycE-Cdk2, mCD8::GFP, gal80ts*) with control flies expressing *lacZ* (*TH>lacZ, mCD8::GFP, gal80ts*). Gene expression was induced by shifting 2-day-old adult flies from 19°C to 29°C. Flies overexpressing CCRs exhibited significantly shorter lifespans compared to controls ($P < 0.0001$, Log-rank Mantel-Cox test). Pale blue and pale pink lines represent individual survival curves from three experimental replicates, while dark blue and red thicker lines indicate the mean survival for the experimental and control groups, respectively. Error bars indicate SD. These findings indicate that forced mitotic re-entry leads to reduced longevity, likely due to apoptosis or mitotic catastrophe in neurons.

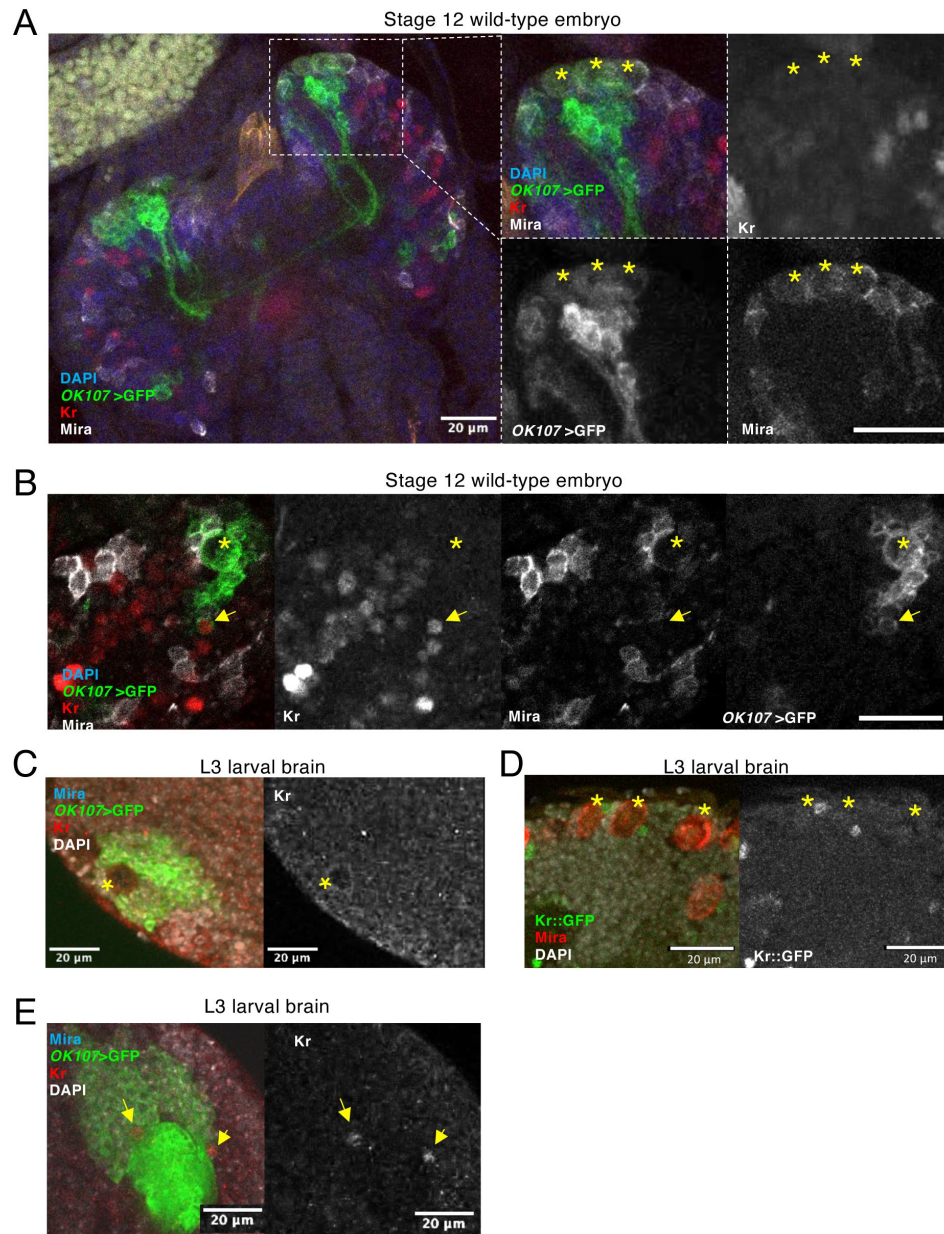


Figure S3.

Kr expression in MBNBs during embryonic and larval stages

(A, B) Representative confocal images of the embryonic central nervous system (CNS) (Stage 12) in wild-type embryos stained with rabbit anti-Kr antibody (red), *OK107>GFP* (green), DAPI (blue), and Mira (grey). (Left panel) Zoomed-out confocal image showing the overall CNS structure. (Right panels) Magnified views of MBNBs, showing merged coloured images and individual grey-scale channels. Yellow asterisks indicate MBNBs identified by *OK107>GFP* and Mira co-expression. (B) Arrows indicate a neuron within the MB lineage expressing Kr. Although Kr was broadly expressed in the embryonic CNS, it was absent from MBNBs. Scale bars: 20 μ m. (C-E) Kr expression in third-instar larval brains. (C) A confocal image of the MB cell body region stained for *OK107>GFP* (green), Mira (cyan), and Kr (red) using a rat anti-Kr antibody, with the individual Kr channel shown in grey on the right. Asterisks mark MBNBs, which exhibit weak Kr signals. (D) *Kr::GFP*(Bac) reporter expression in the MB cell body region, showing *Kr::GFP* (green), DAPI (grey), and Mira (red), with the individual GFP channel shown in grey on the right. Asterisks mark MBNBs. Strong *Kr::GFP* signals were detected in single cells adjacent to MBNBs, likely representing GMCs or early-born neurons, which might inherit *Kr::GFP* from MBNBs. (E) A confocal image of the MB calyx area stained for *OK107>GFP* (green), Mira (cyan), and Kr (red), with the individual Kr channel shown in grey. Kr-expressing neurons near the MB calyx are indicated by arrowheads. Scale bars: 20 μ m.

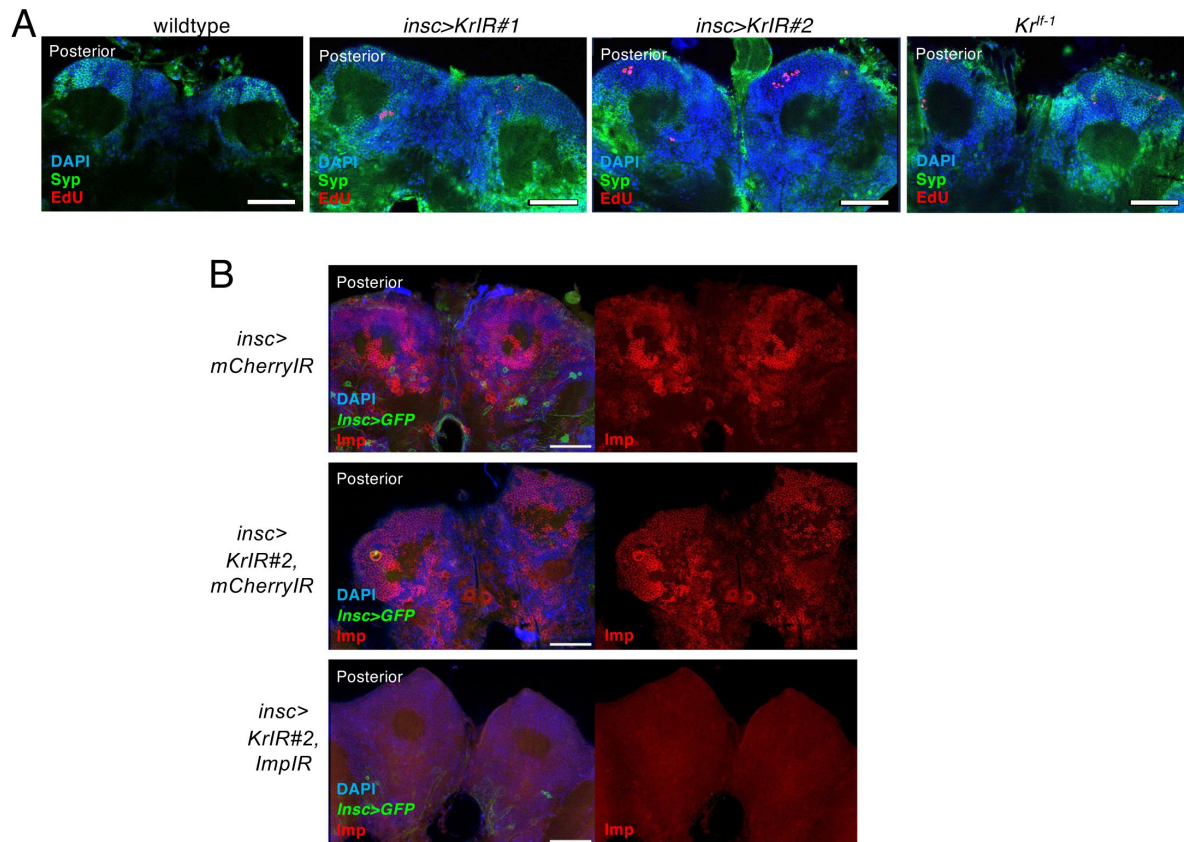


Figure S4.

Syp expression in persisting MBNBs of *Kr* RNAi and *KrIf-1* mutant adult brains.

(A) Representative confocal images of wild type, *KrIf-1* mutant, and *insc>KrIR* adult brains, showing dorsoposterior MB cell body regions. EdU (red) labels proliferating cells, Syp expression (green) is visualised with Syp-specific antibodies, and nuclei are stained with DAPI (blue). Unlike Imp, Syp-expressing regions remained largely unchanged despite Kr depletion. Scale bars: 50 μ m. (B) Imp depletion suppresses the expansion of mitotic MBNBs in Kr RNAi adult brains. (Left) Representative confocal images of control *insc>mCherryIR* (upper panels), *insc>KrIR#2, mCherryIR* (middle panels), and *insc>KrIR#2, ImpIR* (lower panels) brains, immunostained for Imp (red), *insc>GFP* (green), and DAPI (blue). Induction of *Imp* RNAi under *insc*-Gal4 effectively depleted Imp expression in brain tissue including MB lineages. Scale bars: 50 μ m.

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Reviewer #1 (Public review):

Summary:

In this manuscript, the authors investigated factors required for neural progenitors to exit the cell cycle before the adult stage. They first show that Kr is turned on in pupal stage MBNBs, and depletion of Kr from pupal stage NBs leads to retention of MBNBs into the adult stage. Then they demonstrate that these retained NBs maintain the expression of Imp, and co-depletion of Imp abolishes the extended neurogenesis. Further, they show that co-depletion of *kr-h1* significantly reduces the retained MBNBs caused by loss of *kr*, suggesting antagonistic genetic interactions between these two. In addition, they demonstrate that over-expressing *Kr-h1* leads to the striking phenotype of tumor-like neuroblast overgrowth in adult brains.

Strengths:

- (1) The authors leveraged well-controlled, powerful genetic tools (including temporal control of RNAi knockdown using the Gal80ts system), and provided strong evidence that Kr expression in pupal stage MBNBs is required to repress Imp and promote the end of neurogenesis. Similarly, the experimental result of co-depleting *Kr-h1* and *Kr*, and the striking phenotype upon *Kr-h1* mis-expression, support the antagonistic roles played by *Kr-h1* and *Kr* in this process.
- (2) The sample sizes, quantification methods, and p-values are well documented for all experiments. In most parts, the data presented strongly support their conclusions.
- (3) Identification of two transcription factors with opposite roles in controlling cell cycle exit, and their possible interactions with the Imp/Syp axis, is highly significant for the study on how the proliferation of neural progenitors is regulated and limited before the adult stage.

Weaknesses:

(1) The nature of the KrIf-1 allele is not clear. It is mentioned that this allele leads to misexpression of Kr in various tissues. However, it is not clear if Kr is mis-expressed or lost in MBNBs in the KrIf-1 mutant. If Kr is mis-expressed in MBNBs in the KrIf-1 mutant, then it would be difficult to explain why both loss of Kr and mis-expression of Kr in MBNBs lead to the same NB retention phenotype. The authors should examine Kr expression in MBNBs in the KrIf-1 mutant.

(2) Some parts of the regulations and interactions between Kr, Kr-h1, Imp, Syp, and E93 are not well-defined. For example, the data suggest that Kr is turned on in the pupal stage MBNBs, and is required to end neurogenesis through repressing Imp and Kr-h1. To further support this conclusion, the authors can examine if Kr-h1 expression is up-regulated in kr-RNAi. The authors suggested that Kr-h1 may act upstream or in parallel to Imp/Syp, but also suggested that Kr-h1 may repress E93. The expression of Imp, Syp, and E93 can be examined in brains with Kr-h1 mis-expression to determine where Kr-h1 acts. If Imp expression is elevated when Kr-h1 is mis-expressed, then Kr-h1 may act upstream of Imp. If Imp/Syp expression does not change, then Kr-h1 may act on the E93 level.

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Reviewer #2 (Public review):

Summary:

In this paper, the authors study the role of Kruppel in regulating the survival of mushroom body neuroblasts. They first confirm that adult wild-type brains have no proliferation and report that Kruppel mutants and Kruppel RNAi in neuroblasts show a few proliferative clones; they show that these proliferative clones are localized in the mushroom body. They then show that Kruppel is expressed mostly during pupal stages and acts by downregulating the expression of Imp, which has been shown to positively regulate neuroblast proliferation and survival. Expectedly, this also affects neuronal diversity in the mushroom body, which is enriched in gamma neurons that are born during the Imp-expression window. Finally, they show that Kr acts antagonistically to Kr-h1, which is expressed predominantly in larval stages.

Strengths:

The main strength of this paper is that it identified a novel regulator of Imp expression in the mushroom body neuroblasts. Imp is a conserved RNA-binding protein that has been shown to regulate neural stem cell proliferation and survival in different animals.

Weaknesses:

(1) The main weakness of the paper is that the authors want to test adult neurogenesis in a system where no adult neurogenesis exists. To achieve this, they force neuroblasts to survive in adulthood by altering the genetic program that prevents them from terminating their proliferation. If this was reminiscing about "adult neurogenesis", the authors should at least show how adult neurons incorporate into the mushroom body even if they are born much later. On the contrary, this more likely resembles a tumorigenic phenotype, when stem cells divide way past their appropriate timing.

(2) Moreover, the figures are, in many cases, hard to understand, and the interpretation of the figures doesn't always match what one sees. The manuscript would benefit from better figures; for example, in Figure 2C, Miranda expression in *insc>GFP* in Kr-IF-1 is not visible.

(3) The authors describe a targeted genetic screen, but they don't describe which genes were tested, how they were chosen, and why Kruppel was finally selected.

(4) The authors argue that Kr does not behave as a typical tTF in MBNBs. However, they show no expression in the embryo, limited expression in the larva and early pupa, and a peak around P24-P48. This sounds like a temporally regulated expression of a transcription factor. Importantly, they mentioned that they tested their observations against different datasets (FlyAtlas2, modENCODE, and MBNB-lineage-specific RNA-seq data), but they don't provide the data.

(5) Finally, the contribution of Kr to the neuronal composition of the mushroom body is expected (since Imp is known to regulate neuronal diversity in the MB), but the presentation in the paper is very incomplete.

Unfortunately, based on the above, I am not convinced that the authors can use this framework to infer anything about adult neurogenesis. Therefore, the impact of this work is limited to the role of Kruppel in regulating Imp, which has already been shown to regulate the extent of neuroblast division, as well as the neuronal types that are born at different temporal windows.

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Reviewer #3 (Public review):

Summary:

Drosophila neuroblasts (NBs) serve as a well-established model for studying neural stem cell biology. The intrinsic genetic programs that control their mitotic potential throughout development have been described in remarkable detail, highlighting a series of sequentially expressed transcription factors and RNA-binding proteins that together constitute the temporal patterning system.

However, the mechanisms that limit the number of NB divisions remain largely unknown in a specific subset of NBs known as mushroom body neuroblasts (MB NBs). Unlike other NBs, which terminate proliferation before or shortly after the onset of metamorphosis, MB NBs continue dividing until the end of metamorphosis, ceasing only just before adulthood. In this study, the authors identify the transcription factor Krüppel (Kr), a member of the conserved Krüppel-like family, as temporally regulated in MB NBs. They demonstrate that Kr knockdown during pupal stages maintains expression of the RNA-binding protein Imp and results in prolonged MB NB proliferation into adulthood. Their data suggest that Kr contributes to the timely silencing of Imp during metamorphosis. The authors further identify Kr-h1, a related transcription factor, as a potential antagonist. While Kr-h1 appears dispensable for the timely termination of MB NBs under normal conditions, its overexpression leads to their continued proliferation and tumor-like expansion in adults.

This work provides the first evidence for a transcription factor-driven temporal regulation mechanism in MB NBs, offering new insight into the control of neural stem cell self-renewal. Given the evolutionary conservation of Krüppel-like factors, this study may have broader implications for the neural stem cell field.

Strengths:

(1) The study possibly identifies a new series of temporal transcription factors that are specific for mushroom body neuroblasts.

(2) The mechanism could be conserved in vertebrates.

Weaknesses:

Some proposed regulatory interactions, particularly between Kr, Kr-h1, and other temporal factors like Imp, Chinmo, and E93, have not been thoroughly investigated, which weakens the support for the proposed model. Additional experimental validation is needed to confirm these relationships and strengthen the mechanistic framework.

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Author response:

We thank the editors and reviewers for their thoughtful and constructive evaluation of our manuscript, “Krüppel Regulates Cell Cycle Exit and Limits Adult Neurogenesis of Mushroom Body Neural Progenitors in *Drosophila*.” We are pleased that all reviewers recognised the novelty and significance of identifying Krüppel (Kr) as a key transcription factor promoting timely termination of mushroom body neuroblast (MBNB) proliferation, and the potential antagonistic function of Kr-h1.

We appreciate the helpful suggestions aimed at improving the mechanistic clarity and presentation of our findings. Below, we outline how we plan to address the major points raised in the full revision.

(1) Characterisation of the *KrIf-1* allele and Kr expression

We agree that clarifying the nature of the *KrIf-1* allele is important. In response to this concern, we will examine Kr expression in *KrIf-1* mutant larval, pupal, and adult brains using immunostaining and available reporter lines. These experiments will help determine whether the observed neuroblast retention phenotype correlates with altered Kr expression in MBNBs.

(2) Regulatory relationships between Kr, Kr-h1, Imp, Syp, Chinmo, and E93

We are currently performing additional experiments to clarify the interactions among these temporal factors. For instance, we are testing whether Kr-h1 overexpression alters the expression of Imp, Syp, and E93. We have obtained a published E93 antibody from Dr Chris Doe (Syed et al., 2017) and will include E93 expression analysis in our revised manuscript.

While Chinmo is of interest, its expression is well established to be regulated downstream of Imp/Syp via mRNA stability (Liu et al., 2015; Ren et al., 2017). Given that we currently lack reliable tools to assess Chinmo levels, we will focus primarily on Imp, Syp, and E93 as readouts for Kr/Kr-h1 function. If we succeed in obtaining Chinmo antibodies or reporter lines in time, we will include corresponding data.

(3) Expression of Kr-h1 in MBNBs

We fully agree that direct evidence for *Kr-h1* expression in MBNBs is important. To address this, we have obtained the *Kr-h1::GFP* BAC transgenic line (BDSC #96786) and are currently using it to assess *Kr-h1* expression in MBNBs. We also tested an anti-Kr-h1 antibody previously reported by Kang et al. (2017), developed in the context of fat body studies, but it did not yield clear signals in larval MBNBs. However, previous work by Shi et al. (2007) clearly demonstrated Kr-h1 expression in the developing MB, including MBNBs, using a custom antibody developed by their lab. We also contacted the Lee lab to request this antibody, but unfortunately, it is no longer available. We will include the results obtained using the GFP BAC line in the revised manuscript and, if needed, pursue RNA in situ hybridisation to further validate *Kr-h1* expression in MBNBs.

(4) Temporal Kr knockdown and MARCM analysis

We appreciate the suggestion to validate our RNAi-based temporal knockdown results using MARCM. We plan to perform MBNB-specific MARCM analysis following the strategy described by Rossi et al. (2020). However, this approach requires additional time due to the logistics of acquiring the necessary fly stocks, generating appropriate genetic combinations, and conducting clonal analyses. While we will make every effort to include these data, we note that RNAi-based knockdown offers the advantage of temporal reversibility and has been essential for assessing stage-specific requirements in our current study.

(5) Details of the targeted genetic screen

Kr was initially identified as part of a broader, ongoing effort to screen for candidate transcription factors and cell cycle regulators involved in neuroblast cell cycle exit and/or quiescence. As this screen is still preliminary and incomplete, we prefer not to include the full dataset at this stage. Instead, we will revise the manuscript to clarify that *Kr* was prioritised for further investigation based on the striking MBNB-specific phenotype observed upon RNAi-mediated knockdown and in the *KrIf-1* mutant, rather than through a completed screening process.

(6) Clarifying the model (Figure 6D) and interactions

We will revise the proposed model to distinguish between experimentally supported interactions and speculative ones. As noted above, we will primarily focus on the Imp/Syp and E93 axis in relation to Kr and Kr-h1 activity. Chinmo will be omitted from the model unless further data become available to support its inclusion.

(7) Clarifications on figures and data presentation

We appreciate the feedback on figure clarity. We will revise figures such as 1B, 2C, and 3A to improve legibility and presentation. We will also correct typographical errors and figure references, and clarify the activity patterns of the GAL4 drivers. Specifically, while UASmCD8::GFP expression driven by OK107-GAL4 is markedly weaker in MBNBs than in their neuronal progeny (as seen, for example, in Figure S3C), the driver remains active and functionally relevant in MBNBs. We believe the weak expression in MBNBs likely explains the absence of a NB retention phenotype in OK107>KrIR adult brains (see main text, Lines 374–376). As suggested by the reviewer, we will clarify this point earlier in the manuscript and can include additional data showing OK107>GFP expression patterns in pupal MB lineages as supplementary material.

(8) Analysis of public datasets

We will include results from our analysis of publicly available datasets such as FlyAtlas2, modENCODE, and a time-course RNA-seq dataset specific to MBNBs (Liu et al., 2015). While the spatial resolution of FlyAtlas2 and modENCODE is limited, the MBNB dataset provides valuable temporal information up to 36 h after puparium formation (APF). From this dataset, we observe that Kr expression remains consistently low throughout development, with only a modest increase at 84 h ALH (mean TPM ~11) and 36 h APF (~7), suggesting it does not undergo strong transcriptional regulation in MBNBs. In contrast, Kr-h1 is highly expressed during early larval stages (24–84 h ALH; mean TPM ~55–60) and shows a marked suppression by 36 h APF (mean TPM ~2), consistent with its proposed role in promoting MBNB proliferation. Importantly, Eip93F (E93) exhibits a reciprocal pattern to Kr-h1—with minimal expression until 84 h ALH (mean TPM ~24), followed by a substantial induction at 36 h APF (mean TPM ~104), aligning with its known role in triggering neuroblast termination. These temporal expression dynamics support our model that Kr-h1 and E93 function in opposition during the transition from proliferative to terminating neuroblast states. We will summarise

these findings in the revised manuscript, along with appropriate discussion of dataset limitations.

We hope this provisional response conveys our strong commitment to thoroughly addressing the reviewers' concerns and improving the manuscript. We are currently carrying out additional experiments and will submit a revised version with new data and enhanced clarity in due course.

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